ABSTRACT

Crime analysis and prediction using machine learning aims to identify crime patterns and forecast potential criminal activities based on historical data. This project utilizes real- world crime datasets containing attributes such as city, victim demographics, weapon used, and police deployment to analyze crime trends. Data preprocessing and feature selection techniques are applied to enhance data quality and model efficiency. Multiple machine learning algorithms are trained and evaluated to achieve accurate crime classification and prediction. The trained model predicts the probable type of crime for given inputs through a user-friendly web interface developed using Flask. Visualization techniques such as graphs, accuracy charts, and confusion matrices are used to interpret results effectively. The system assists law enforcement agencies in identifying crime- prone areas and understanding crime behavior. This project demonstrates how machine learning can support proactive crime prevention and decision-making for public safety.This project utilizes real- world crime datasets containing attributes such as city, victim demographics, weapon used, and police deployment to analyze crime trends.Day by day crime data rate is increasing because the modern technologies and hi-tech methods are helping the criminals to achieving the illegal activities. According to Crime Record Bureau crimes like burglary, arson etc., have been increased while crimes like murder, sex, abuse, gang rap etc., have been increased. Crime data will be collected from various blogs, news and websites. It also helps to see if a crime in a certain known pattern or a new pattern necessary. Crimes can be predicted as the criminal are active and operate in their comfort Zone.

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LIST OF ABBREVIATIONS

NLP	Natural Language Processing
SVM	Support Vector Machine
NER	Named Entity Recognition
RNN	Recurrent Neural Networks
PDS	Project Developed System
CNN	Convolutional Neural Network

CHAPTER - 01

INTRODUCTION

1.1 PROJECT OVERVIEW

Crime is one of the major challenges affecting the safety, economic growth, and social stability of society. With the rapid increase in crime data, traditional analysis methods have become inefficient and time-consuming. Machine learning provides advanced techniques to analyze large volumes of crime data and discover hidden patterns. This project focuses on analyzing and predicting crime rates using real-world datasets. By applying data preprocessing and feature selection, meaningful information is extracted from raw crime records. Various machine learning models are trained to classify and predict crime types accurately. A web-based interface is developed to allow users to input data and obtain predictions in real time. Visualization techniques are used to represent crime trends and patterns effectively. The system helps identify crime-prone areas and time periods. This approach supports law enforcement agencies in making informed decisions to reduce crime.

Day by day crime data rate is increasing because the modern technologies and hi-tech methods are helping the criminals to achieving the illegal activities. According to Crime Record Bureau crimes like burglary, arson etc., have been increased while crimes like murder, sex, abuse, gang rap etc., have been increased. Crime data will be collected from various blogs, news and websites. The huge data is used as a record for creating a crime report database. The knowledge which is acquired from the data mining techniques will help in reducing crimes as it helps in finding the culprits faster and also the areas that are most affected by crime. Data mining helps in solving the crimes faster and this technique gives good results when applied on crime dataset, the information obtained from the data mining techniques can help the police department. A particular approach has been found to be useful by the police, which is the identification of crime 'hot spots 'which indicates areas with a high concentration of crime. Use of data mining techniques can produce important results from crime report datasets.

1.2 PROBLEM DEFINITION

Design and implement a comprehensive system for Crime Rate Analysis and Prediction that leverages historical crime data, socio-economic factors, and geographical information to provide actionable insights for law enforcement agencies and policymakers. The goal is to develop a predictive model capable of identifying trends, hotspots, and potential areas at risk of criminal activities, enabling proactive measures to be taken to prevent and address crime. The system should not only analyze past crime data but also incorporate real-time information and evolving socio-economic conditions to enhance the accuracy and timeliness of predictions. Additionally, the solution should prioritize user- friendly interfaces for easy interpretation of results and facilitate efficient decision-making for law enforcement and relevant stakeholders.

1.3 OBJECTIVES

The objectives of our work are to:

- Predicting crime before it takes place.
- Predicting hotspots of crime.
- Understanding crime pattern.
- Classify crime based on location.

CHAPTER - 02

LITERATURE SURVEY

1) An Exploration of Crime Prediction Using Data Mining on Open Data

AUTHORS: **Ginger Saltos and Minhaela Cocea (2017)**

The increase in crime data recording coupled with data analytics resulted in the growth of research approaches aimed at extracting knowledge from crime records to better understand criminal behavior and ultimately prevent future crimes. While many of these approaches make use of clustering and association rule mining techniques, there are fewer approaches focusing on predictive models of crime. In this paper, we explore models for predicting the frequency of several types of crimes by LSOA code (Lower Layer Super Output Areas — an administrative system of areas used by the UK police) and the frequency of anti-social behavior crimes. Three algorithms are used from different categories of approaches: instance-based learning, regression and decision trees. The data are from the UK police and contain over 600,000 records before preprocessing. The results, looking at predictive performance as well as processing time, indicate that decision trees (M5P algorithm) can be used to reliably predict crime frequency in general as well as anti-social behavior frequency. The experiments were conducted using the SCIAMA High Performance Computer Cluster at the University of Portsmouth.

2) Crime Analysis and Prediction Using Data Mining

AUTHORS: Shiju Sathyadevan, Devan M.S, Surya Gangadharan (IEEE-2014)

Crime analysis and prevention is a systematic approach for identifying and analyzing patterns and trends in crime. Our system can predict regions which have high probability for crime occurrence and can visualize crime prone areas. With

the increasing advent of computerized systems, crime data analysts can help the Law enforcement officers to speed up the process of solving crimes. Using the concept of data mining we can extract previously unknown, useful information from an unstructured data. Here we have approach between computer science and criminal justice to develop a data mining procedure that can help solve crimes faster. Instead of focusing on causes of crime occurrence like criminal background of offender, political enmity etc. we are focusing mainly on crime factors of each day. This paper has tested the accuracy of classification and prediction based on different test sets. Classification is done based on the Bayes theorem which showed more than 90% accuracy.It also helps to see if a crime in a certain known pattern or a new pattern necessary.Crimes can be predicted as the criminal are active and operate in their comfort Zones.

3) Crime Detection Techniques Using Data Mining and K-Means

AUTHORS: Khushabu A.Bokde, Tisksha P. Kakade, Dnyanes hwari S. Tumsare, Chetan G. Wadhai (2018)

Crimes will somehow influence organizations and institutions when occurred frequently in a society. Thus, it seems necessary to study reasons, factors and relations between occurrence of different crimes and finding the most appropriate ways to control and avoid more crimes. The main objective of this paper is to classify clustered crimes based on occurrence frequency during different years. Data mining is used extensively in terms of analysis, investigation and discovery of patterns for occurrence of different crimes.

4) Crime Pattern Analysis, Visualization and Prediction Using Data Mining

AUTHORS: Rajkumar. S, Sakkarai , Soundarya Jagan. J, Varnikasree. P (2015)

Crime against women these days has become problem of every nation around the globe many countries are trying to curb this problem. Preventive are taken to reduce the increasing number of cases of crime against women. A huge amount of data set is generated every year on the basis of reporting of crime. This data can prove very useful in analyzing and predicting crime and help us prevent the crime to some extent. Crime analysis is an area of vital importance in police department. Study of crime data can help us analyze crime pattern, inter-related clues& important hidden relations between the crimes. That is why data mining can be great aid to analyze, visualize and predict crime using crime data set. Classification and correlation of data set makes it easy to understand similarities & dissimilarities amongst the data objects. We group data objects using clustering technique. Dataset is classified on the basis of some predefined condition. Here grouping is done according to various types of crimes against women taking place in different states and cities of India. Crime mapping will help the administration to plan strategies for prevention of crime, further using data mining technique data can be predicted and visualized in various form in order to provide better understanding of crime patterns.

We applied a theoretical model based on data mining techniques such as clustering and classification to real crime dataset recorded by police in England and Wales within 1990 to 2011. We assigned weights to the features in order to improve the quality of the model and remove low value of them. The Genetic Algorithm (GA) is used for optimizing of Outlier Detection operator parameters using Rapid Miner tool.The spread of fake news by social bots.

5) Survey paper on crime prediction using ensemble approach

AUTHORS: Ayisheshim Almaw, Kalyani kadam (2018)

Crime is a foremost problem where the top priority has been concerned by individual, the community and government. This paper investigates a number of data mining algorithms and ensemble learning which are applied on crime data mining. This survey paper describes a summary of the methods and techniques which are implemented in crime data analysis and prediction. Crime forecasting is a way of trying to mining out and decreasing the upcoming crimes by forecasting the future crime that will occur. Crime prediction practices historical data and after examining data, predict the upcoming crime with respect to location, time, day, season and year. In present crime cases rapidly increases so it is an inspiring task to foresee upcoming crimes closely with better accuracy. Data mining methods are too important to resolving crime problem with investigating hidden crime patterns.so the objective of this study could be analyzing and discussing various methods which are applied on crime prediction and analysis. This paper delivers reasonable investigation of Data mining Techniques and ensemble classification techniques for discovery and prediction of upcoming crime.

6) Empirical Analysis for Crime Prediction and Forecasting Using Machine Learning and Deep Learning Techniques

AUTHORS: Wajiha Safat, Sohail Asghar, Saira Andleeb Gillani (IEEE-2021)

Crime and violation are the threat to justice and meant to be controlled. Accurate crime prediction and future forecasting trends can assist to enhance metropolitan safety computationally. The limited ability of humans to process complex information from big data hinders the early and accurate prediction and forecasting of crime. The accurate estimation of the crime rate, types and hot spots from past patterns creates many computational challenges and opportunities. Despite considerable research efforts, yet there is a need to have a better predictive algorithm, which direct police patrols toward criminal activities. Previous studies are lacking to achieve crime.

7) Survey on crime analysis and predict ion using data mining techniques

AUTHORS: Benjamin Fredrick David. H and Suruliand I (2017)

Data Mining is the procedure which includes evaluating and examining large pre-existing databases in order to generate new information which may be essential to the organization. The extraction of new information is predicted using the existing datasets. Many approaches for analysis and prediction in data mining had been performed. But many few efforts have made in the criminology field. Many few have taken efforts for comparing the information all these approaches produce. The police stations and other similar criminal justice agencies hold many large databases of information which can be used to predict or analyze the criminal movements and criminal activity involvement in the society. The criminals can also be predicted based on the crime data. The main aim of this work is to perform a survey on the supervised learning and unsupervised learning techniques that has been applied towards criminal identification. This paper presents the survey on the Crime analysis and crime prediction using several Data Mining techniques. The quantitative analysis produced results which shows the increase in the Accuracy level of classification because of using the GA to optimize the parameters.

Therefore, this study applied different machine learning algorithms, namely, the logistic regression, support vector machine (SVM), Naïve Bayes, k-nearest neighbors (KNN), decision tree, multilayer perceptron (MLP), random forest, and extreme Gradient Boosting, and time series analysis by long short-term memory (LSTM) and autoregressive integrated moving average (ARIMA) model to better fit the crime data. The performance of LSTM for time series analysis was reasonably adequate in order of magnitude of root mean square error (RMSE) and mean absolute error (MAE), on both data sets. Exploratory data analysis predicts more than 35 crime types and overall, these results provide early identification of crime, hot spots with higher crime rate.

8) Systematic Literature Review of Crime Prediction and Data Mining

AUTHORS: Falade Adesola and Ambrose Azeta (2019)

Using crime datasets requires different strategies for the varying types of data that describe illicit activity. Falade et al. (2019) provide a survey of crime prediction efforts wherein various machine learning methods have been applied to multiple types of datasets: criminal records, social media, news, and police reports. The authors note the different opportunities and challenges that each type of crime dataset presents, such as social media posts being highly unstructured and First Information Reports (FIRs) being unstructured but reliable. This paper is explaining techniques used, challenges addressed, methodologies used, and crime data mining and analysis paper. The methodologies is composed of three stages the first stage involves the research work related to crime data mining, second stage is concerned with establishing a classification and the third stage is involves the presentation of summary of research in crime data mining and analysis.

Thus, it seems necessary to study reasons, factors and relations between occurrence of different crimes and finding the most appropriate ways to control and avoid more crimes. The main objective of this paper is to classify clustered crimes based on occurrence frequency during different years. Data mining is used extensively in terms of analysis, investigation and discovery of patterns for occurrence of different crimes.

CHAPTER - 3

SYSTEM ANALYSIS

3.1. EXISTING SYSTEM

Data mining in the study and analysis of criminology can be categorized into main areas, crime control and crime suppression. De Bruin et. Al. introduced a framework for crime trends using a new distance measure for comparing all individuals based on their profiles and then clustering them accordingly. Manish Gupta et. Al. highlights the existing systems used by Indian police as e-governance initiatives and also proposes an interactive query-based interface as crime analysis tool to assist police in their activities. He proposed interface which is used to extract useful information from the vast crime database maintained by National Crime Record Bureau (NCRB) and find crime hot spots using crime data mining techniques such as clustering etc. The effectiveness of the proposed interface has been illustrated on Indian crime records. Sutapat Thiprungsri examines the application of cluster analysis in the accounting domain, particularly discrepancy detection in audit. The purpose of his study is to examine the use of clustering technology to automate fraud filtering during an audit. He used cluster analysis to help auditors focus their efforts when evaluating group life insurance claims.

3.1.1 Disadvantages of Existing System

- Increase in crime information that has to be stored and analyzed.
- Analysis of data is difficult since data is incomplete and inconsistent.
- Limitation in getting crime data records from the Law Enforcement department

3.2. PROPOSED SYSTEM

In this project, we will be using the technique of machine learning and data science for crime prediction of crime data set. The crime data is extracted from

the official portal of police. It consists of crime information like location description, type of crime, date, time, latitude, longitude. Before training of the model data preprocessing will be done following this feature selection and scaling will be done so that accuracy

Obtain will be high. The Logistic Regression classification and various other algorithms (Decision Tree and Random Forest) will be tested for crime prediction and one with better accuracy will be used for training. Visualization of dataset will be done in terms of graphical representation of many cases for example at which time the criminal rates are high or at which month the criminal activities are high. The whole purpose of this project is to give a just idea of how machine learning can be used by the law enforcement agencies to detect, predict and solve crimes at a much faster rate and thus reduces the crime rate. This can be used in other states or countries depending upon the availability of the dataset.

3.2.1. Advantages of Proposed System

- Identifying the crime hotspots.
- Uncover the crime patterns.
- Predicting the future crime occurrences.

CHAPTER - 04

SYSTEM SPECIFICATIONS

4.1. HARDWARE REQUIREMENTS:

- **System** : Intel Core i5
- **Hard Disk** : 512 GB
- **Ram** : 8 GB

4.2. SOFTWARE REQUIREMENTS:

- **Operating system** : Windows 10
- **Coding Language** : Python
- **Front-End** : HTML,CSS
- **IDE** : Visual Studio Code

4.3 SOFTWARE DESCRIPTION

Python:

Python is a free, open-source programming language. Therefore, all you have to do is install Python once, and you can start working with it. Not to mention that you can contribute own code to the community. Python is also a cross-platform compatible language. So, what does this mean? Well, you can install and run Python on several operating systems. Whether you have a Windows, Mac or Linux, you can rest assure that Python will work on all these operating systems. Python is also a great visualization tool. It provides libraries such as Matplotlib, seaborn and bokeh to create stunning visualizations.

Python is the most popular language for machine learning and deep learning. As a matter of fact, today, all top organizations are investing in Python to implement machine learning in the back-end. Python is a general-purpose interpreted, interactive, object-oriented, and high-level programming language. It was created by Guido van Rossum during 1985- 1990. Like Perl, Python source code is also available under the GNU General Public License (GPL). It was developed by Guido van Rossum in the late 80's and early 90's at the National Research Institute for Mathematics and Computer Science in the Netherlands. Python is also known as Object Oriented Programming language.

Python is derived from many other languages, including ABC, Modula-3, C, C++, Algol-68, SmallTalk, and Unix shell and other scripting languages. It is copyrighted Like Perl; Python source code is now available under the GNU General Public License (GPL). It is now maintained by a core development team at the institute, although Guido van Rossum still holds a vital role in directing its progress.

Applications Of Python:

- **Easy-to-learn** – Python has few keywords, simple structure, and a clearly defined syntax. This allows the student to pick up the language quickly.
- **Easy-to-read** – Python code is more clearly defined and visible to the eyes.
- **Easy-to-maintain** – Python's source code is fairly easy-to-maintain.
- **A broad standard library** – Python's bulk of the library is very portable and cross- platform compatible on UNIX, Windows, and Macintosh.
- **Interactive Mode** – Python has support for an interactive mode which allows interactive testing and debugging of snippets of code.
- **Portable** – Python can run on a wide variety of hardware platforms and has the same interface on all platforms.
- **Extendable** – You can add low-level modules to the Python interpreter. These

modules enable programmers to add to or customize their tools to be more efficient.

- **Databases** − Python provides interfaces to all major commercial databases.
- **GUI Programming** − Python supports GUI applications that can be created and ported to many system calls, libraries and windows systems, such as Windows MFC, Macintosh, and the X Window system of Unix.
- **Scalable** − Python provides a better structure and support for large programs than shell scripting.

Features Of Python:

- It supports functional and structured programming methods as well as OOP.
- It can be used as a scripting language or can be compiled to byte-code for building large applications.
- It provides very high-level dynamic data types and supports dynamic type checking
- It supports automatic garbage collection.
- It can be easily integrated with C, C++, COM, ActiveX, CORBA, and Java.

Machine Learning:

Machine learning, a captivating field within artificial intelligence (AI), empowers machines to learn from data and improve their performance over time. Just like humans learn from experiences, machine learning algorithms analyze vast amounts of data to uncover patterns, identify relationships, and make predictions without explicit programming. This remarkable ability has revolutionized numerous domains, from healthcare and finance to self-driving cars and recommendation systems.

Machine Learning Algorithms can be classified into 3 types as follows: -

- Supervised Learning
- Unsupervised Learning
- Reinforcement Learning

Supervised Learning

Supervised learning is defined as when a model gets trained on a "Labeled Dataset". Labeled datasets have both input and output parameters. In Supervised Learning algorithms learn to map points between inputs and correct outputs. It has both training and validation datasets labeled.

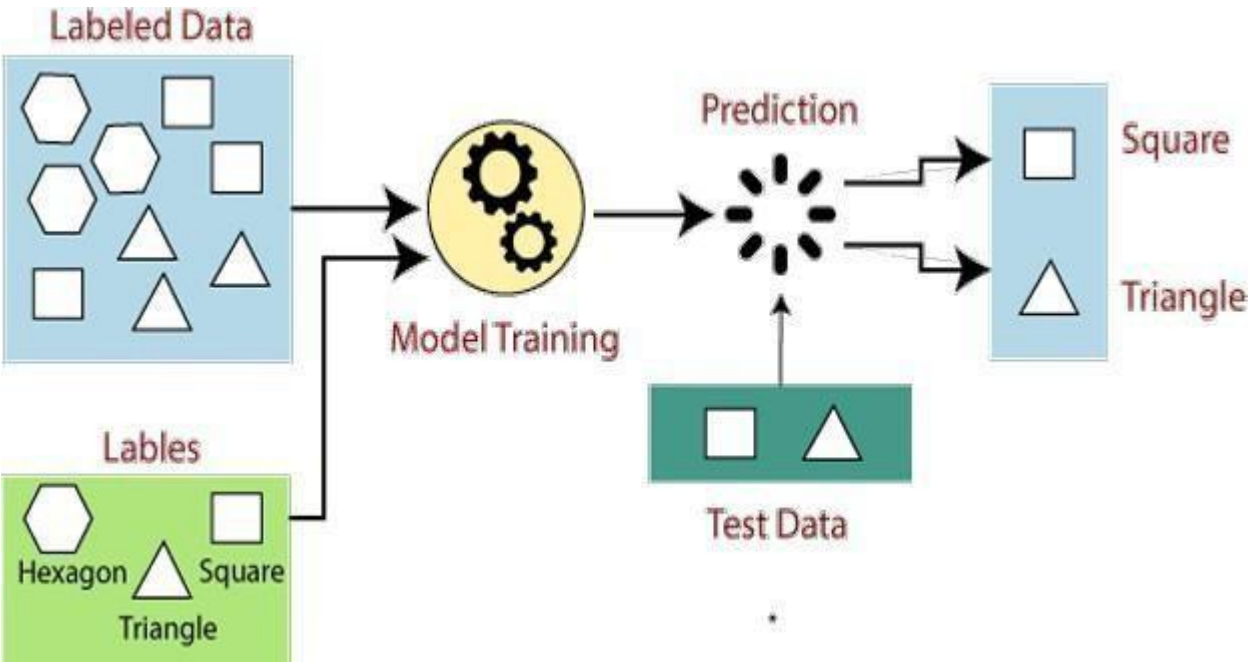


Figure 4.1: - Supervised Learning

Unsupervised Learning:

Unlike supervised learning, unsupervised learning deals with unlabeled data, where the data points lack predefined categories. The objective here is to uncover hidden patterns within the data itself. This is commonly used in tasks like image segmentation, where the algorithm automatically groups pixels into meaningful objects or regions within an image.

Reinforcement Learning:

This branch draws inspiration from how humans learn through trial and error. The learning process involves an agent interacting with an environment and receiving

rewards for desirable actions and penalties for undesirable ones. Over time, the agent learns to maximize its rewards through exploration and exploitation, making it suitable for applications like game playing and robot control.

Artificial intelligence (AI), empowers machines to learn from data and improve their performance over time. Just like humans learn from experiences, machine learning algorithms analyze vast amounts of data to uncover patterns, identify relationships, and make predictions without explicit programming. This remarkable ability has revolutionized numerous domains, from healthcare and finance to self-driving cars and recommendation systems.

So, what does this mean? Well, you can install and run Python on several operating systems. Whether you have a Windows, Mac or Linux, you can rest assure that Python will work on all these operating systems. Python is also a great visualization tool. It provides libraries such as Matplotlib, seaborn and bokeh to create stunning visualizations

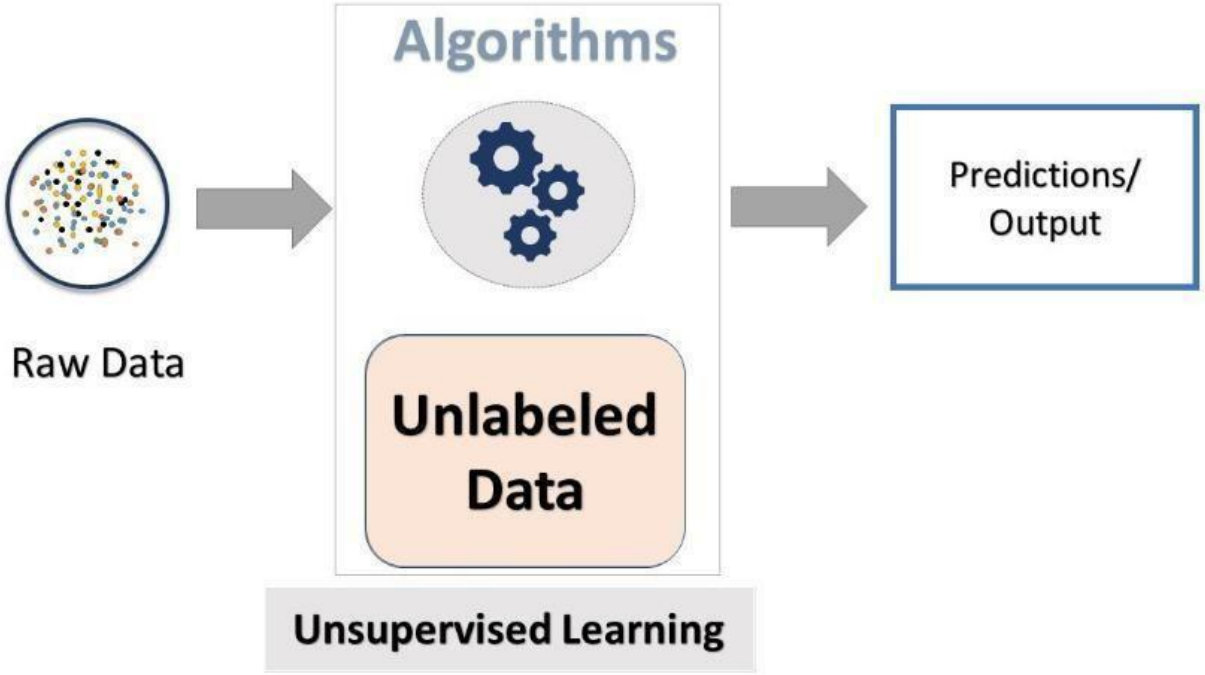


Figure 4.2: - Unsupervised Learning

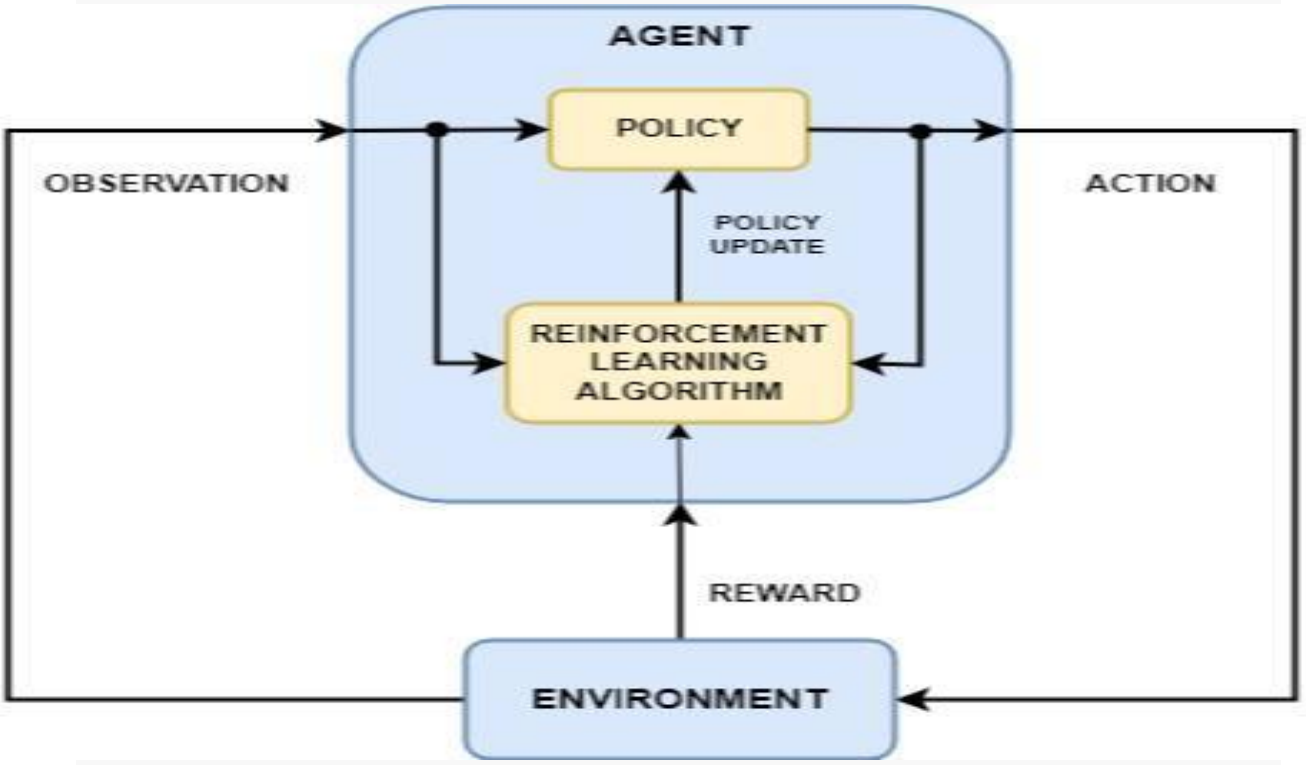


Figure 4.3: - Reinforcement Learning

CHAPTER - 05

5.1. SYSTEM ARCHITECTURE

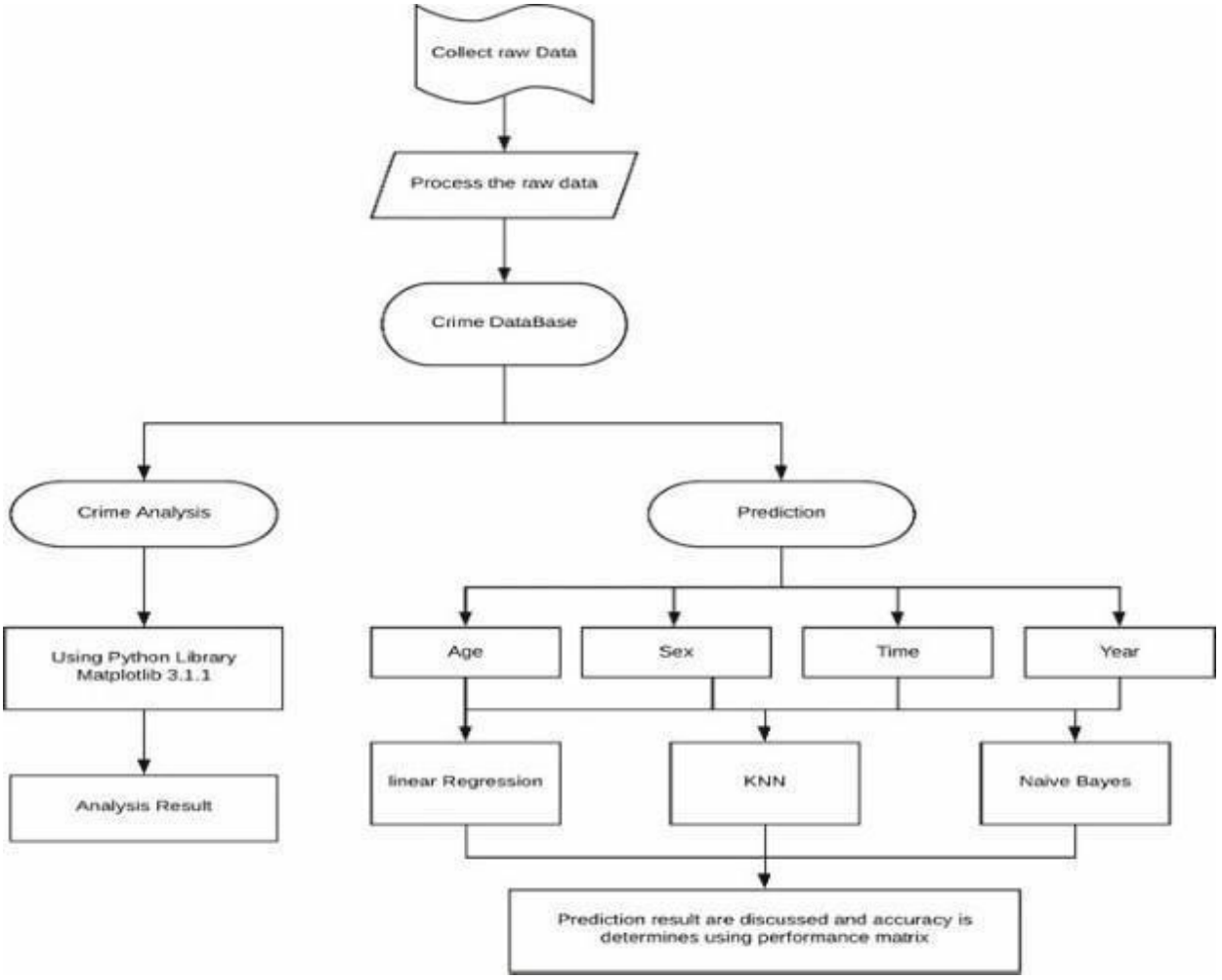


Figure5.1: - System Architecture

For system developers, they need system architecture diagrams to understand, clarify, and communicate ideas about the system structure and the user requirements that the system must support.

It describes the overall features of the software is concerned with defining the requirements and establishing the high level of the system. During architectural design, the various web pages and their interconnections are identified and designed. The major software components are identified and decomposed into processing modules and conceptual data structures and the interconnections among the modules are identified. The following modules are identified in the proposed system.

The system architectural design is the design process for identifying the subsystems making up the system and framework for subsystem control and communication. The goal of the architectural design is to establish the overall structure of software system.

5.2. DATA FLOW DIAGRAM

- The DFD is also called as bubble chart. It is a simple graphical formalism that can be used to represent a system in terms of input data to the system, various processing carried out on this data, and the output data is generated by this system.

- The data flow diagram (DFD) is one of the most important modeling tools. It is used to model the system components. These components are the system process, the data used by the process, an external entity that interacts with the system and the information flows in the system.

DFD shows how the information moves through the system and how it is modified by a series of transformations. It is a graphical technique that depicts information flow.

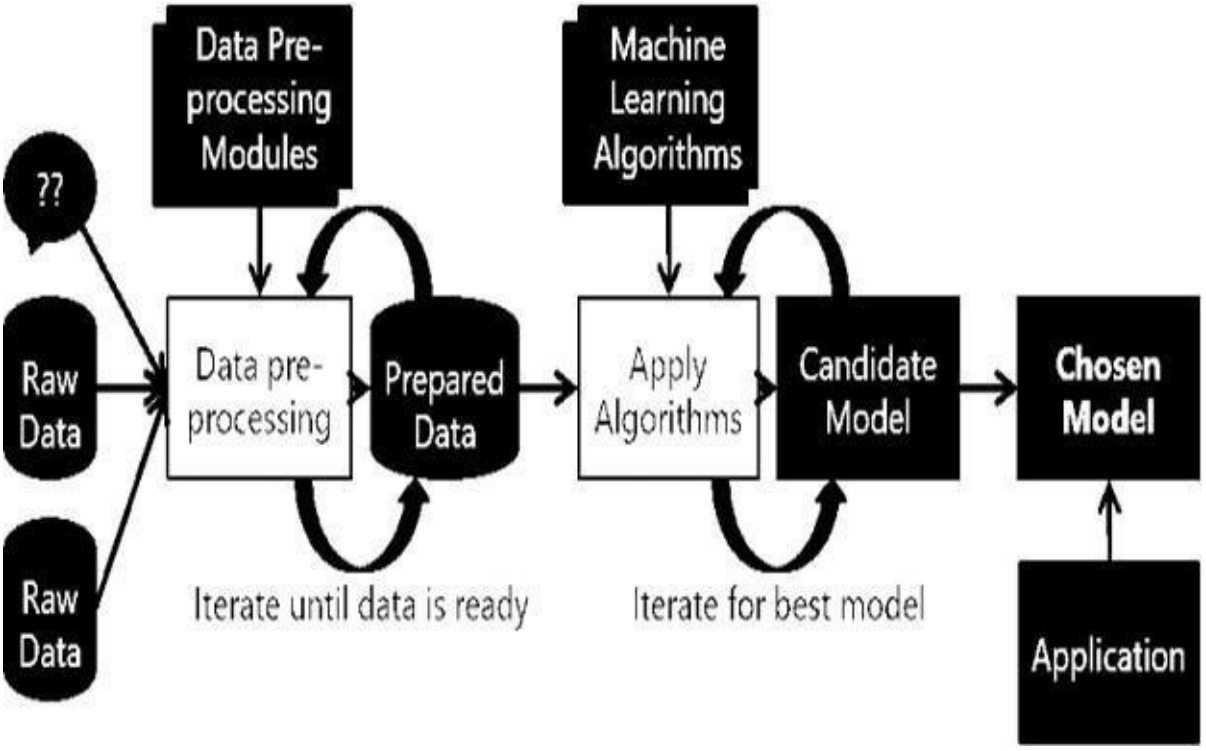


Fig-5.2 Data Flow Diagram

5.3. UML DIAGRAMS

UML stands for Unified Modeling Language. UML is a standardized general-purpose modeling language in the field of object-oriented software engineering. The standard is managed, and was created by, the Object Management Group.

The goal is for UML to become a common language for creating models of object-oriented computer software. In its current form UML is comprised of two major components: a Meta-model and a notation. In the future, some form of method or process may also be added to; or associated with, UML.

The Unified Modeling Language is a standard language for specifying, Visualization, Constructing and documenting the artifacts of software system, as well as for business modeling and other non-software systems.The UML represents a collection of best engineering practices that have proven successful in the modeling of large and complex systems.

The UML is a very important part of developing objects-oriented software and the software development process. The UML uses mostly graphical notations to express the design of software projects.

GOALS:

The Primary goals in the design of the UML are as follows:

1. Provide users a ready-to-use, expressive visual modeling Language so that they can develop and exchange meaningful models.
2. Provide extendibility and specialization mechanisms to extend the core concepts.
3. Be independent of particular programming languages and development process.
4. Provide a formal basis for understanding the modeling language.
5. Encourage the growth of OO tools market.
6. Support higher level development concepts such as collaborations, frameworks, patterns and components.

5.4. CLASS DIAGRAM:

In software engineering, a class diagram in the Unified Modeling Language (UML)

is a type of static structure diagram that describes the structure of a system by showing the system's classes, their attributes, operations (or methods), and the relationships among the classes. It explains which class contains information.

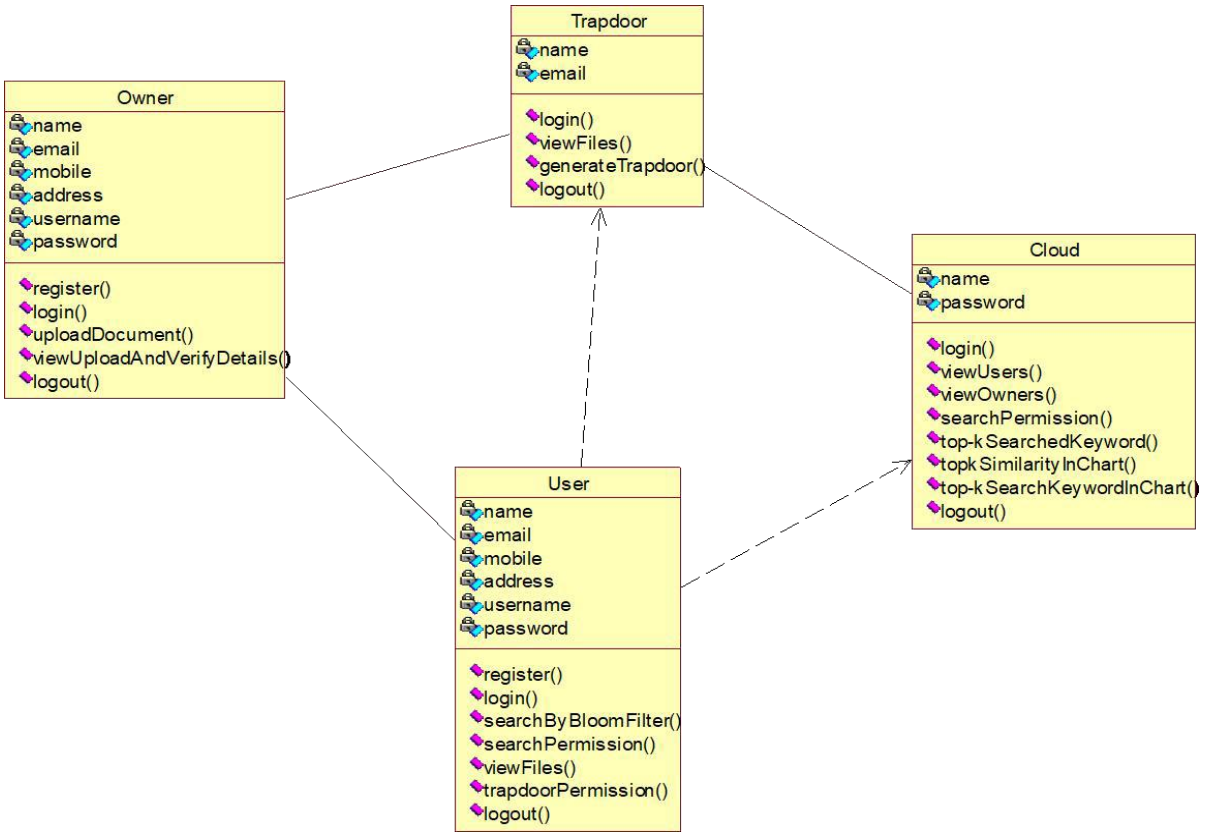


Fig.5.4.1: Class diagram of the project

5.5. SEQUENCE DIAGRAM:

A sequence diagram in Unified Modeling Language (UML) is a kind of interaction diagram that shows how processes operate with one another and in what order. It is a construct of a Message Sequence Chart. Sequence diagrams are sometimes called event diagrams, event scenarios, and timing diagrams.

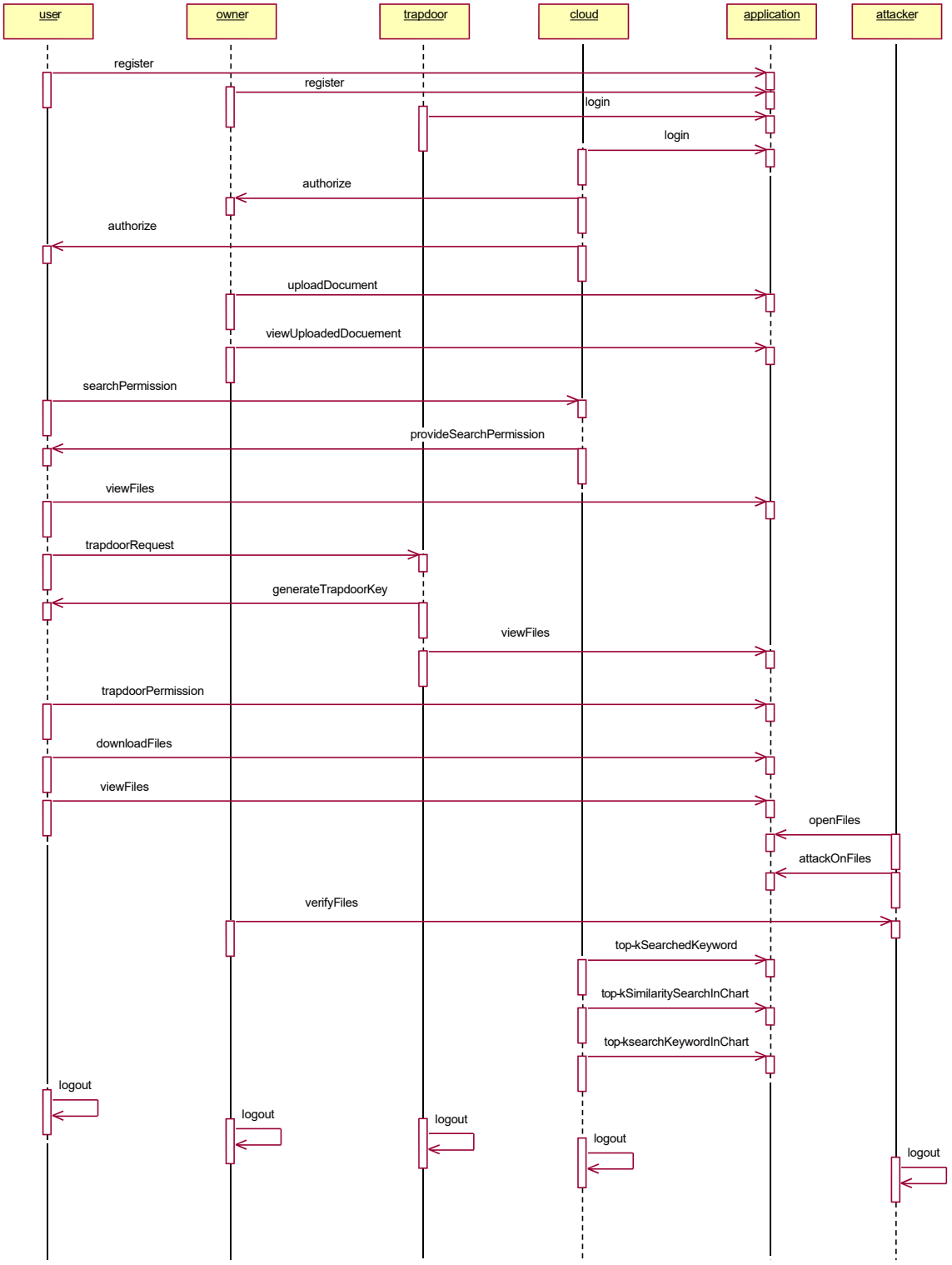


Fig.5.5.1: Sequence Daigaram

5.6. USE CASE DIAGRAM:

A use case diagram in the Unified Modeling Language (UML) is a type of behavioral diagram defined by and created from a Use-case analysis. Its purpose is to present a graphical overview of the functionality provided by a system in terms of actors, their goals (represented as use cases), and any dependencies between those use cases. The main purpose of a use case diagram is to show what system functions are performed for which actor. Roles of the actors in the system can be depicted.

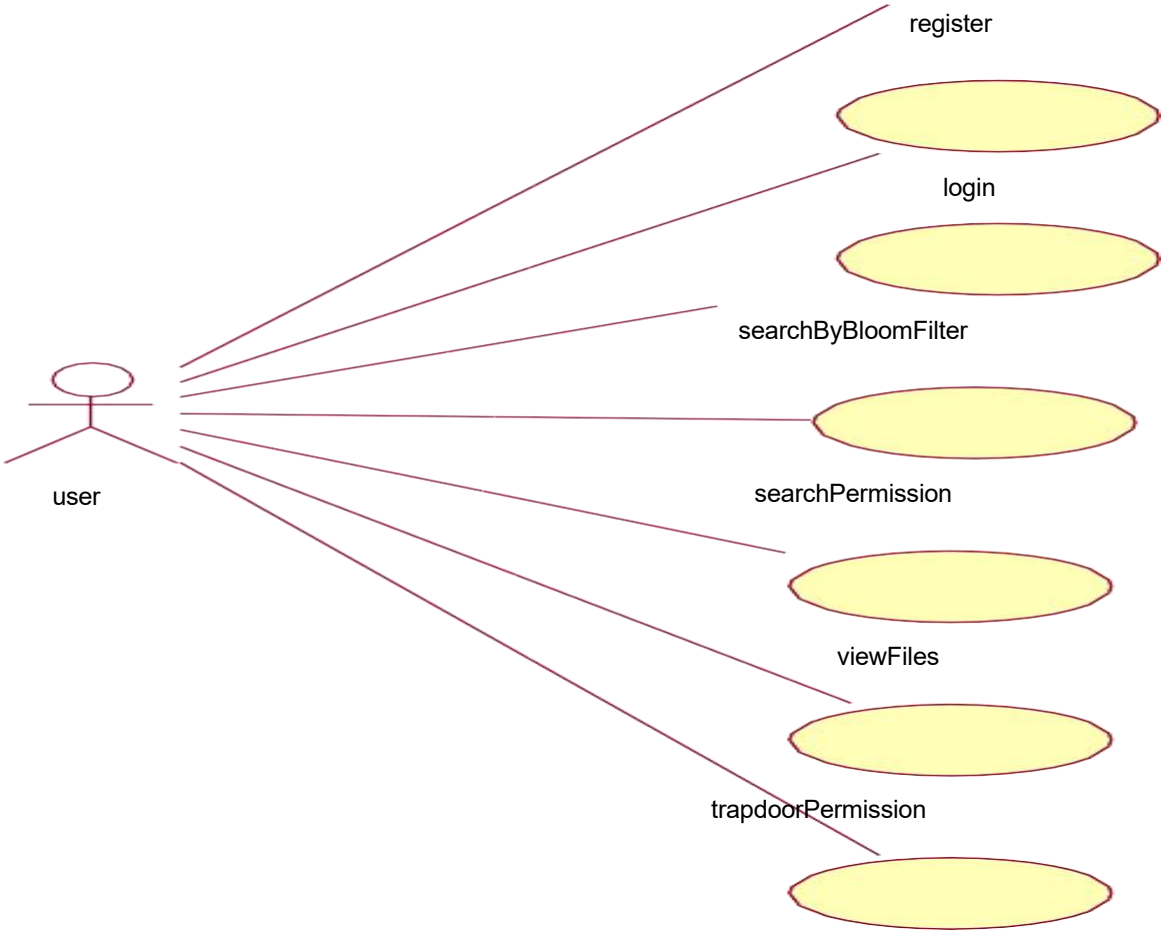


Fig.5.6.1: Use Case Daigram

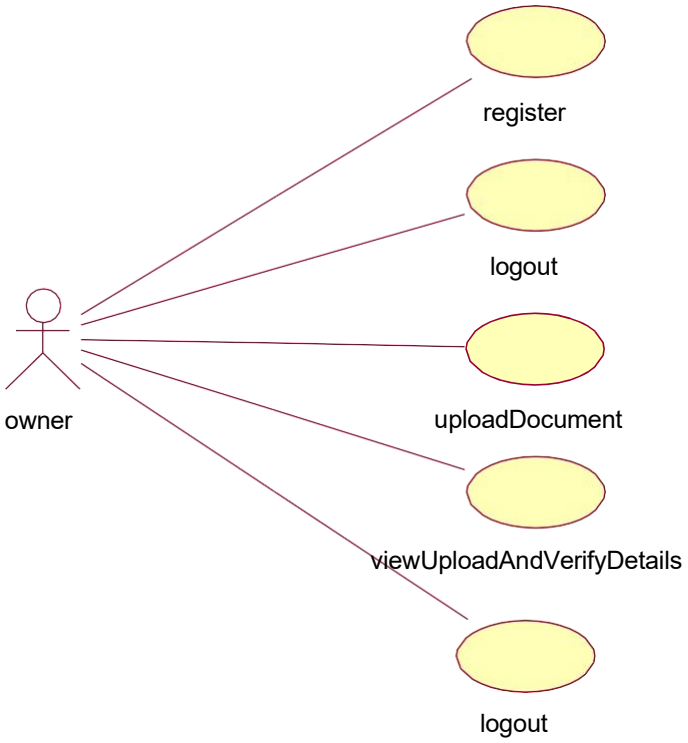


Fig.5.6.2: Use case diagram

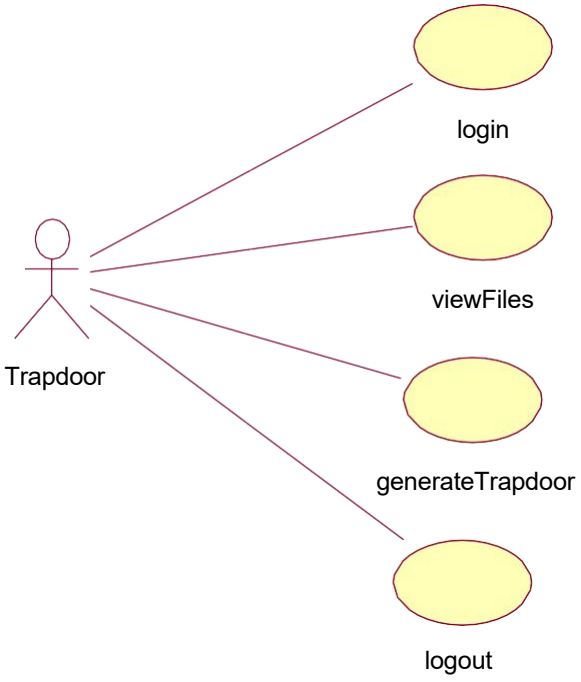


Fig.5.6.3: Use case diagram

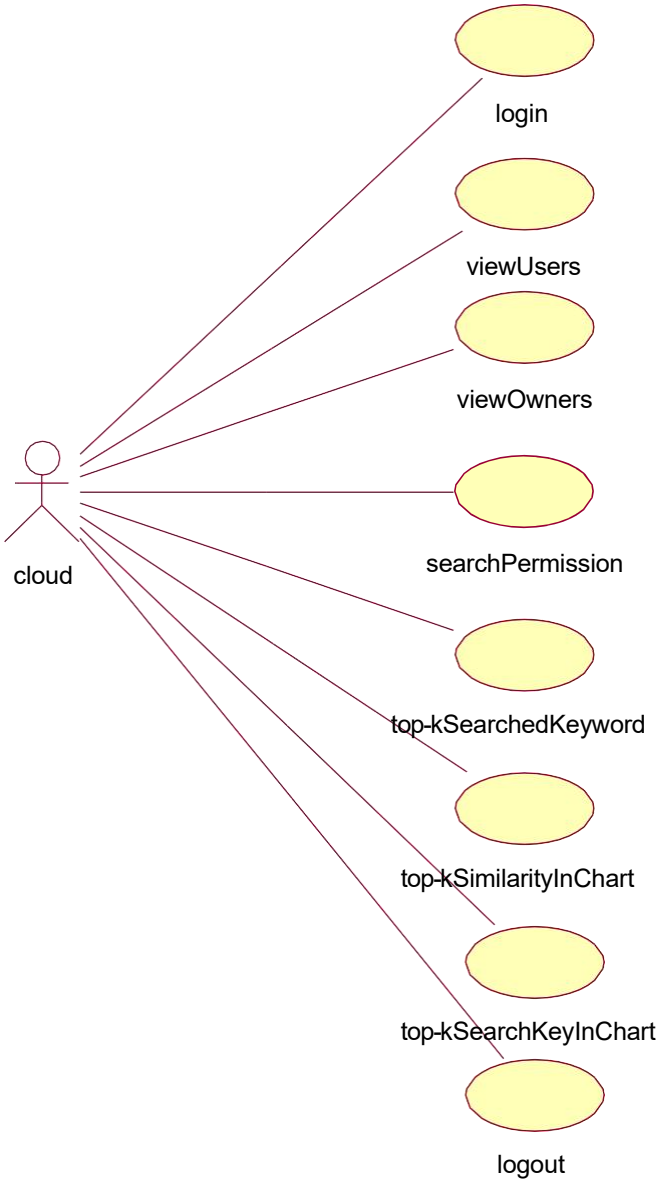


Fig.5.6.4: Use case diagram

CHAPTER-6

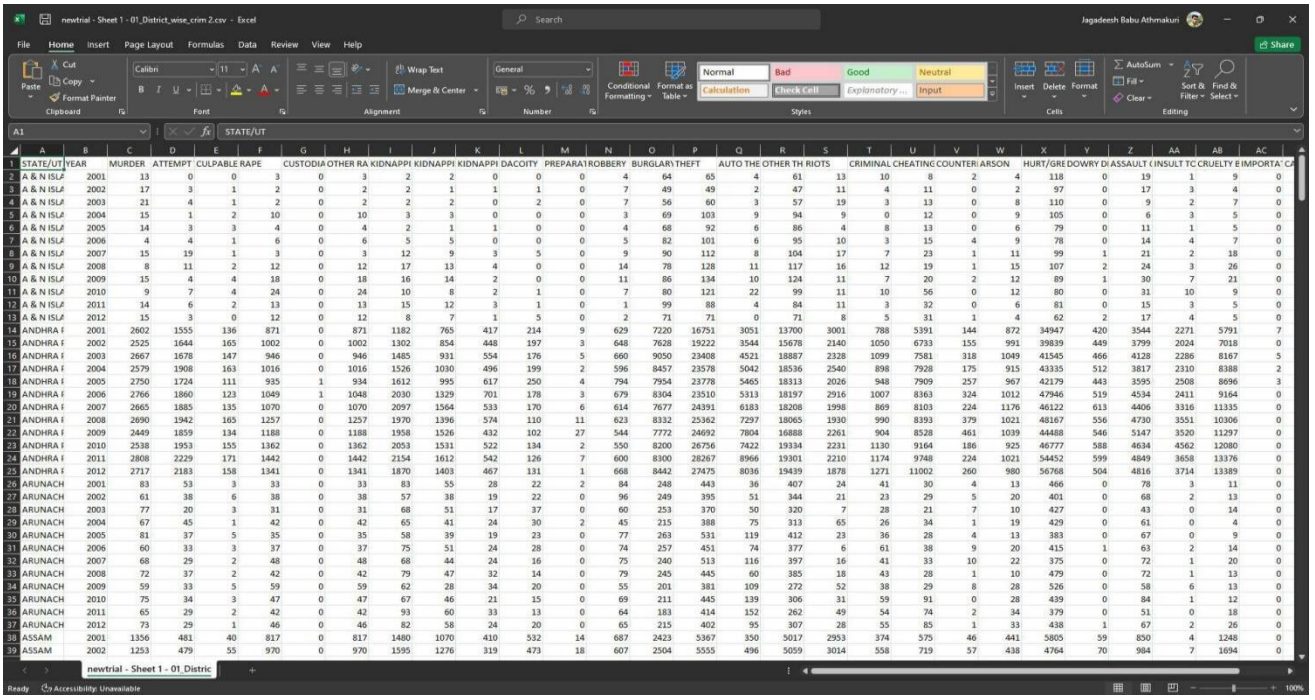
SYSTEM IMPLEMENTATION

6.1. MODULES

1. Data Collection Module
2. Data Pre-processing Module
3. Feature Selection Module
4. Model Traning Module
5. Prediction Module
6. Visualization Module

6.1.1 Data Collection Module:

The First step of the project is to get the data set “Crime Data set India” from Kaggle It is having 46,160 records with various columns. The data set contains information such as age,gender,police deployment,type of crime,and weapon used.



STATE/UT	YEAR	MURDER	ATTEMPT	CULPABLE	RAPE	CUSTODIA	OTHER RA	KIDNAPPI	KIDNAPPI	KIDNAPPI	DACOITY	PREPARA	ROBBERY	BURGLAR	THEFT	AUTO THE	OTHER TH	RIOTS	CRIMINAL	CHEATING	COUNTERI	ARSON	HURT/GRE	DOWRY DI	ASSAULT (	INSULT TO	CRUELTY B	IMPORTA CA
A & N ISLA	2001	13	0	0	3	0	3	2	2	0	0	0	4	64	65	4	61	13	10	8	2	4	118	0	19	1	9	0
A & N ISLA	2002	17	3	1	2	0	2	2	1	1	1	0	7	49	49	2	47	11	4	11	0	2	97	0	17	3	4	0
A & N ISLA	2003	21	4	1	2	0	2	2	2	0	2	0	7	56	60	3	57	19	3	13	0	8	110	0	9	2	7	0
A & N ISLA	2004	15	1	2	10	0	10	3	3	0	0	0	3	69	103	9	94	9	0	12	0	9	105	0	6	3	5	0
A & N ISLA	2005	14	3	3	4	0	4	2	1	1	0	0	4	68	92	6	86	4	8	13	0	6	79	0	11	1	5	0
A & N ISLA	2006	4	4	1	6	0	6	5	5	0	0	0	5	82	101	6	95	10	3	15	4	9	78	0	14	4	7	0
A & N ISLA	2007	15	19	1	3	0	3	12	9	3	5	0	9	90	112	8	104	17	7	23	1	11	99	1	21	2	18	0
A & N ISLA	2008	8	11	2	12	0	12	17	13	4	0	0	14	78	128	11	117	16	12	19	1	15	107	2	24	3	26	0
A & N ISLA	2009	15	4	4	18	0	18	16	14	2	0	0	11	86	134	10	124	11	7	20	2	12	89	1	30	7	21	0
A & N ISLA	2010	9	7	4	24	0	24	10	8	2	1	0	7	80	121	22	99	11	10	56	0	12	80	0	31	10	9	0
A & N ISLA	2011	14	6	2	13	0	13	15	12	3	1	0	1	99	88	4	84	11	3	32	0	6	81	0	15	3	5	0
A & N ISLA	2012	15	3	0	12	0	12	8	7	1	5	0	2	71	71	0	71	8	5	31	1	4	62	2	17	4	5	0
ANDHRA	2001	2602	1555	136	871	0	871	1182	765	417	214	9	629	7220	16751	3051	13700	3001	788	5391	144	872	34947	420	3544	2271	5791	7
ANDHRA	2002	2525	1644	165	1002	0	1002	1302	854	448	197	3	648	7628	19222	3544	15678	2140	1050	6733	155	991	39839	449	3799	2024	7018	0
ANDHRA	2003	2667	1678	147	946	0	946	1485	931	554	176	5	660	9050	23408	4521	18887	2328	1099	7581	318	1049	41545	466	4128	2286	8167	5
ANDHRA	2004	2579	1908	163	1016	0	1016	1526	1030	496	199	2	596	8457	23578	5042	18536	2540	898	7928	175	915	43335	512	3817	2310	8388	2
ANDHRA	2005	2750	1724	111	935	1	934	1612	995	617	250	4	794	7954	23778	5465	18313	2026	948	7909	257	967	42179	443	3595	2508	8696	3
ANDHRA	2006	2766	1860	123	1049	1	1048	2030	1329	701	178	3	679	8304	23510	5313	18197	2916	1007	8363	324	1012	47946	519	4534	2411	9164	0
ANDHRA	2007	2665	1885	135	1070	0	1070	2097	1564	533	170	6	614	7677	24391	6183	18208	1998	869	8103	224	1176	46122	613	4406	3316	11335	0
ANDHRA	2008	2690	1942	165	1257	0	1257	1970	1396	574	110	11	623	8332	25362	7297	18065	1930	990	8393	379	1021	48167	556	4730	3551	10306	0
ANDHRA	2009	2449	1859	134	1188	0	1188	1958	1526	432	102	27	544	7772	24692	7804	16888	2261	904	8528	461	1039	44488	546	5147	3520	11297	0
ANDHRA	2010	2538	1953	155	1362	0	1362	2053	1531	522	134	2	550	8200	26756	7422	19334	2231	1130	9164	186	925	46777	588	4634	4562	12080	0
ANDHRA	2011	2808	2229	171	1442	0	1442	2154	1612	542	126	7	600	8300	28267	8966	19301	2210	1174	9748	224	1021	54452	599	4849	3658	13376	0
ANDHRA	2012	2717	2183	158	1341	0	1341	1870	1403	467	131	1	668	8442	27475	8036	19439	1878	1271	11002	260	980	56768	504	4816	3714	13389	0
ARUNACH	2001	83	53	3	33	0	33	83	55	28	22	2	84	248	443	36	407	24	41	30	4	13	466	0	78	3	11	0
ARUNACH	2002	61	38	6	38	0	38	57	38	19	22	0	96	249	395	51	344	21	23	29	5	20	401	0	68	2	13	0
ARUNACH	2003	77	20	3	31	0	31	68	51	17	37	0	60	253	370	50	320	7	28	21	7	10	427	0	43	0	14	0
ARUNACH	2004	67	45	1	42	0	42	65	41	24	30	2	45	215	388	75	313	65	26	34	1	19	429	0	61	0	4	0
ARUNACH	2005	81	37	5	35	0	35	58	39	19	23	0	77	263	531	119	412	23	36	28	4	13	383	0	67	0	9	0
ARUNACH	2006	60	33	3	37	0	37	75	51	24	28	0	74	257	451	74	377	6	61	38	9	20	415	1	63	2	14	0
ARUNACH	2007	68	29	2	48	0	48	68	44	24	16	0	75	240	513	116	397	16	41	33	10	22	375	0	72	1	20	0
ARUNACH	2008	72	37	2	42	0	42	79	47	32	14	0	79	245	445	60	385	18	43	28	1	10	479	0	72	1	13	0
ARUNACH	2009	59	33	5	59	0	59	62	28	34	20	0	55	201	381	109	272	52	38	29	8	28	526	0	58	6	13	0
ARUNACH	2010	75	34	3	47	0	47	67	46	21	15	0	69	211	445	139	306	31	59	91	0	28	439	0	84	1	12	0
ARUNACH	2011	65	29	2	42	0	42	93	60	33	13	0	64	183	414	152	262	49	54	74	2	34	379	0	51	0	18	0
ARUNACH	2012	73	29	1	46	0	46	82	58	24	20	0	65	215	402	95	307	28	55	85	1	33	438	1	67	2	26	0
ASSAM	2001	1356	481	40	817	0	817	1480	1070	410	532	14	687	2423	5367	350	5017	2953	374	575	46	441	5805	59	850	4	1248	0
ASSAM	2002	1253	479	55	970	0	970	1595	1276	319	473	18	607	2504	5555	496	5059	3014	558	719	57	438	4764	70	984	7	1694	0

Fig.6.1.1: Crime dataset india having

6.1.2. Data Preprocessing Module:

In this module, the raw data is cleaned and prepared,there are 46,160 entries are present in the data set. We handle missing vales.remove duplicate data,and convert categorical data(like city,gender) into numerical form using label Encoder.

6.1.3. Feature Selection Module:

Here, we select only the important attributes from the dataset.Features selection is done with city,age,gender,weapon used and police deployment are chosen, and the unnecessary data is removed.

6.1.4. Model Training:

In this module ,Machine learning algorithms are applied.here we used:

- Logistic Regression
- Decision Tree
- Random Forest

The dataset is split into training and testing data, and models are trained and evaluated.

6.1.5. Prediction Module:

The trained model is used to predict crime type. User inputs data through a Flask web interface,and the system gives output instantly.The accuracy is calculated using accuracy_score imported from metrics - metrics.accuracy_score (y_test, predicted).

6.1.6. Visualization Module:

This module displays the result in graphical form. Charts.graphs and confusion matrices are used to show crime trends and model accuracy.

6.2 SYSTEM STUDY

Feasibility Study The feasibility of the project is analyzed in this phase and business proposal is put forth with a very general plan for the project and some cost estimates. During system analysis the feasibility study of the proposed system is to be carried out. This is to ensure that the proposed system is not a burden to the company. For feasibility analysis, some understanding of the major requirements for the system is essential.

Three key considerations involved in the feasibility analysis are

- Economic Feasibility
- Technical Feasibility
- Social Feasibility

6.2.1. Economic Feasibility:

This study is carried out to check the economic impact that the system will have on the organization. The amount of fund that the company can pour into the research and development of the system is limited. The expenditures must be justified. Thus, the developed system as well within the budget and this was achieved because most of the technologies used are freely available. Only the customized products had to be purchased.

6.2.2. Technical Feasibility:

This study is carried out to check the technical feasibility, that is, the technical requirements of the system. Any system developed must not have a high demand on the available technical resources. This will lead to high demands on the available technical resources. This will lead to high demands being placed on the client. The developed system must have a modest requirement, as only minimal or null changes are required for implementing this system.

6.2.3. Social Feasibility:

The aspect of study is to check the level of acceptance of the system by the user. This includes the process of training the user to use the system efficiently. The user must not feel threatened by the system, instead must accept it as a necessity. The level of acceptance by the users solely depends on the methods that are employed to educate the user about the system and to make him familiar with it. His level of confidence must be raised so that he is also able to make some constructive criticism, which is welcomed, as he is the final user of the system.

CHAPTER-07

RESULTS AND DISCUSSIONS

7.1. INTRODUCTION

Crime is a major issue affecting the safety and development of society. With the increasing population and urbanization, crime rates are also rising, making it difficult for law enforcement agencies to analyze and control criminal activities using traditional methods.

This project focuses on **crime rate analysis and prediction using machine learning techniques**. By using real-world crime data, the system identifies patterns and trends in criminal activities. Machine learning algorithms such as Logistic Regression, Decision Tree, and Random Forest are used to train models for accurate prediction of crime types.

The project also involves data pre-processing and feature selection to improve the quality of data and model performance. A web-based interface is developed using Flask, allowing users to input data and get predictions in real time.This system helps in identifying crime-prone areas and supports law enforcement agencies in taking preventive measures, there by improving public safety.

7.2. CATEGORIZED CRIME ANALYSIS

Categorized crime analysis refers to the process of organizing crime data into different categories and analyzing each group to understand patterns and trends. In this project, crime data is classified based on factors such as type of crime (theft, robbery, assault, etc.), location (city-wise), demographic details (age and gender), weapon used, and police deployment. By dividing the data into these categories, it becomes easier to identify which crimes occur more frequently, which areas are more prone to crime, and which groups are most affected. This analysis helps in uncovering hidden patterns in the dataset and improves the performance of machine learning models used for prediction. It also supports law .

Table 7.1. Categorized crime analysis types

General Category	Analysis type	Freq.	Percentage	Total
Crime	Neighbourhood	13	43%	52
	Crime	32		
	Crime & Neighbourhood	5		
	Crime density	2		
Spatial	Spatial clustering	3	16%	19
	Spatial correlation	7		
	Hot spot analysis	8		
	Census demographic analysis	2		
Human behaviour analysis	Human behaviour analysis	2	7%	9
	Criminal behaviour analysis	1		
	Facial expression analysis	1		
	Activity analysis	3		
	POI interest	1		
	Surveillance	1		
Other analysis (convergence, qualitative, frequent pattern, feature selection)	Feature selection	3	8%	9
	Predictive policing	1		
	Qualitative	2		
	Convergence	1		
	Frequent pattern	2		
Social media	Object/image analysis	1	6%	7
	Tweets	3		
	Text	1		
	Social Media	2		
Not applicable	NA	24	20%	24
Total				**120**

Table2 presents the frequency of crime types studied in the collected research papers. As shown in the table, robbery is the most studied crime type, followed by murder/homicides and burglary with 51, 41, and 39 papers each, respectively. Most of the research papers studied more than one crime type. For instance, the research paper A100, covered the following crime types: homicide, kidnapping, narcotics, theft, robbery, sexual assault, assault, and use of weapons.

Table 7.2. Frequency of the categories of crimes studied by researchers

Category of Crime	**Freq.**	**Category of Crime**	**Freq.**	**Category of Crime**	**Freq.**
Robbery	51	Fraud	5	Carnapping	2
Murder/Homicides	41	Arson	5	Trafficking	3
Burglary	39	Cybercrime/Hacking	4	Disorder	3
Theft	35	Disorder	3	Attack	1
Assault	35	Vandalism	3	Mascare	1
Vehicle Larceny	18	Injury	3	Children offense	1
Rape	17	Stalking	3	Graffiti crimes	1
Drugs/Narcotics	15	Gambling	2	Dowry death	1
Violence	11	Homeless crimes	2	Cattle rusting	1
Kidnapping	11	Foreclosed	2	Physical injury	1
Sexual assault	11	Dacoits	2	Fighting	1
Gun/weapons/Shooting	7	Accidents	2	Attack	1

7.2. EVALUATION AND PERFORMANCE METRICS

In this project, evaluation metrics are used to analyze the performance of machine learning models such as Logistic Regression, Decision Tree, and Random Forest. Accuracy measures how many predictions are correct overall, while precision and recall help in understanding the model's performance in identifying specific crime categories. The F1-score provides a balance between precision and recall. A confusion matrix is used to visualize the number of correct and incorrect predictions. These metrics help in selecting the best model for accurate crime prediction.

Table 7.3. Performance metric categories

Performance Metric -Main Category	**Performance Metric - Specific types**	**Freq.**	**%**
Mean Error	Mean Absolute Percentage Error (MAPE) Mean Absolute Error (MAE) Mean Squared Error (MSE) Root Mean Squared Error (RMSE) Mean Relative Error (MRE) Error rate	**59**	**21%**
Accuracy	Directional accuracy (DA) Predictive Accuracy Index (PAI) Classification accuracy	**45**	**16%**
True Positive Rate (TPR)	Sensitivity Detection Rate Recall unweighted average recall (UAR) Hit rate Correct Finds Correctly Classified	**38**	**14%**
F-measure	Macro-F1 Micro-F1	**26**	**9%**
Precision	Positive Prediction Value	**26**	**9%**
Performance Metric -Main Category	**Performance Metric - Specific types**	**Freq.**	**%**
Area Under Curve (AUC)	Receiver Operating Characteristic (ROC) Area Under Receiver Operating Characteristic (AUROC)	**20**	**7%**

Correlation coefficient	Pearson Correlation Coefficient (PCC) coefficient of determination (R^2) Rank Correlation Correlation Matrix	**9**	**3%**
False Positive Rate (FPR)	False Alarm Rate (FAR) Correct exclusion	**9**	**3%**
Time	Elapsed Time Detection Time Delay	**7**	**3%**
Confusion matrix	NA	**5**	**1%**
True Negative Rate (TNR)	Specificity false hits	**4**	**1%**
False Negative Rate (FNR)	Misclassified Incorrect exclusions	**4**	**1%**
prediction index	Predictive Efficiency Index (PEI) Recapture Rate Index (RRI) Adjusted Rand Index (ARI)	**4**	**1%**
PAI (Personal Activity Intelligence)	NA	**3**	**1%**
Kappa Statistic	NA	**2**	**0%**
Memory usage	NA	**1**	**0%**
p-value	NA	**1**	**0%**
Jensen-Shannon Divergence (JS)	NA	**1**	**7%**
No applied Performance metric	NA	**19**	**21%**

8.SCREEN SHOTS:

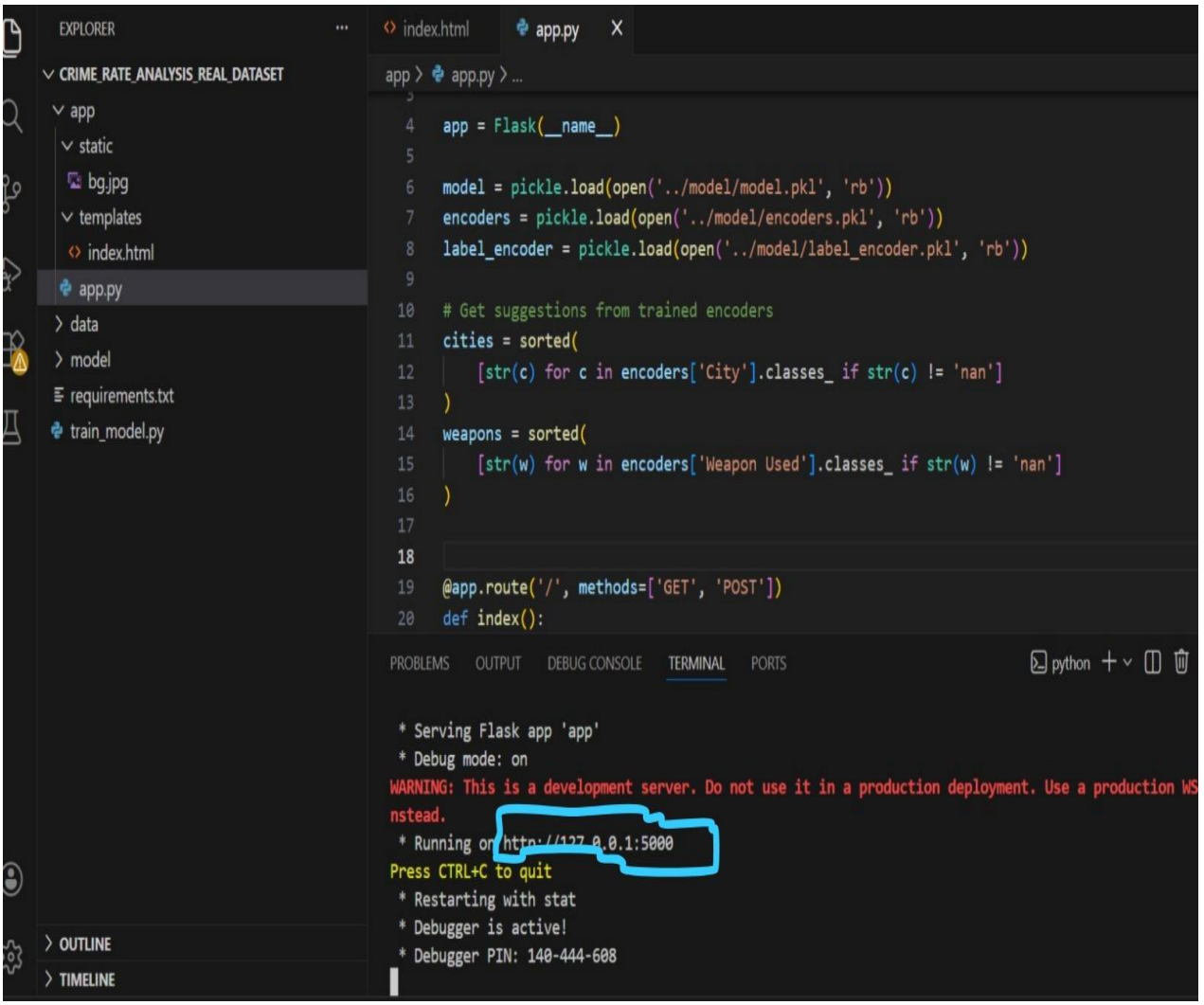


Fig.8.1: Shows the page of running our application

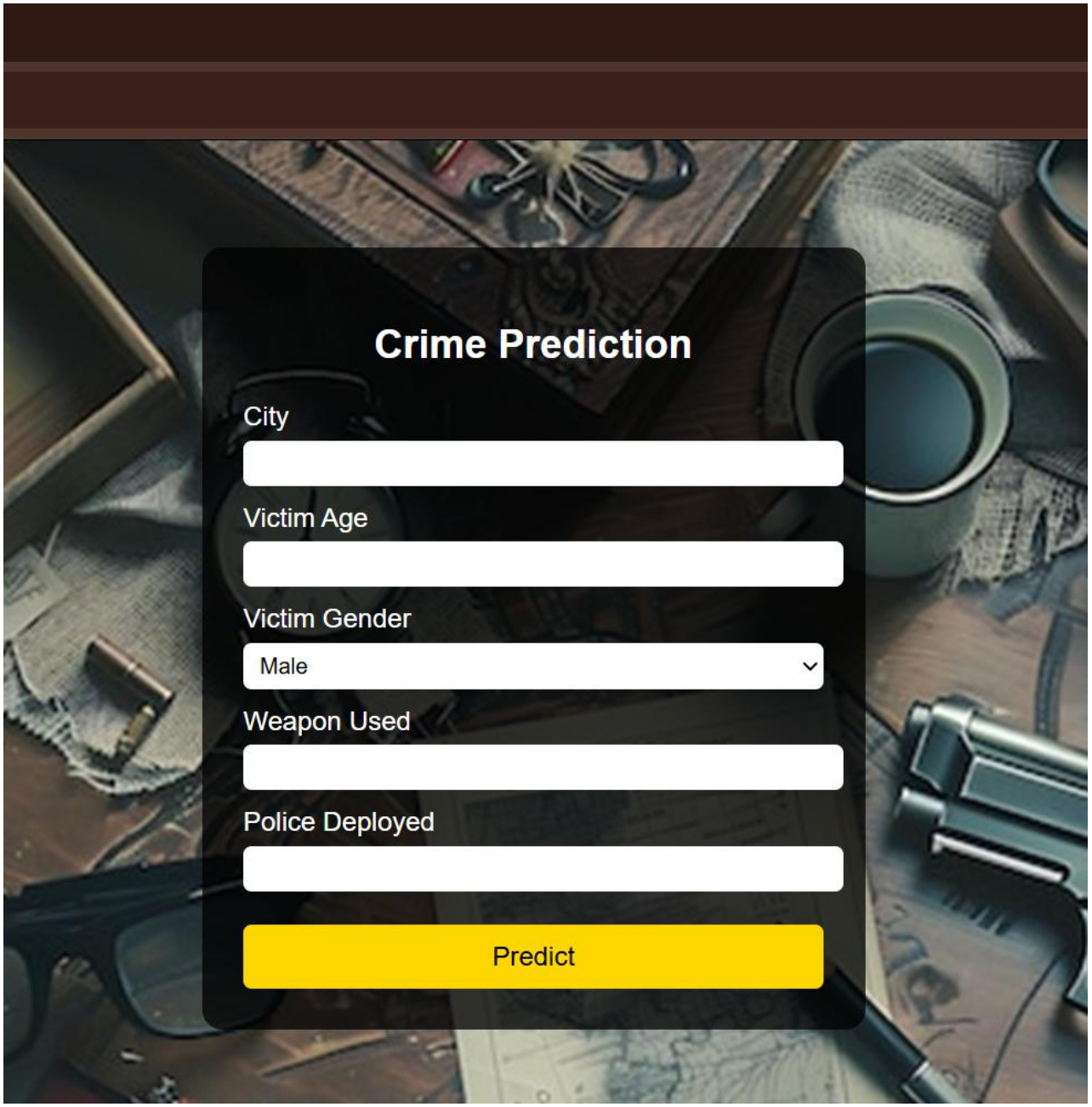


Fig.8.2: Shows the homepage of our application of Crime analysis and prediction.

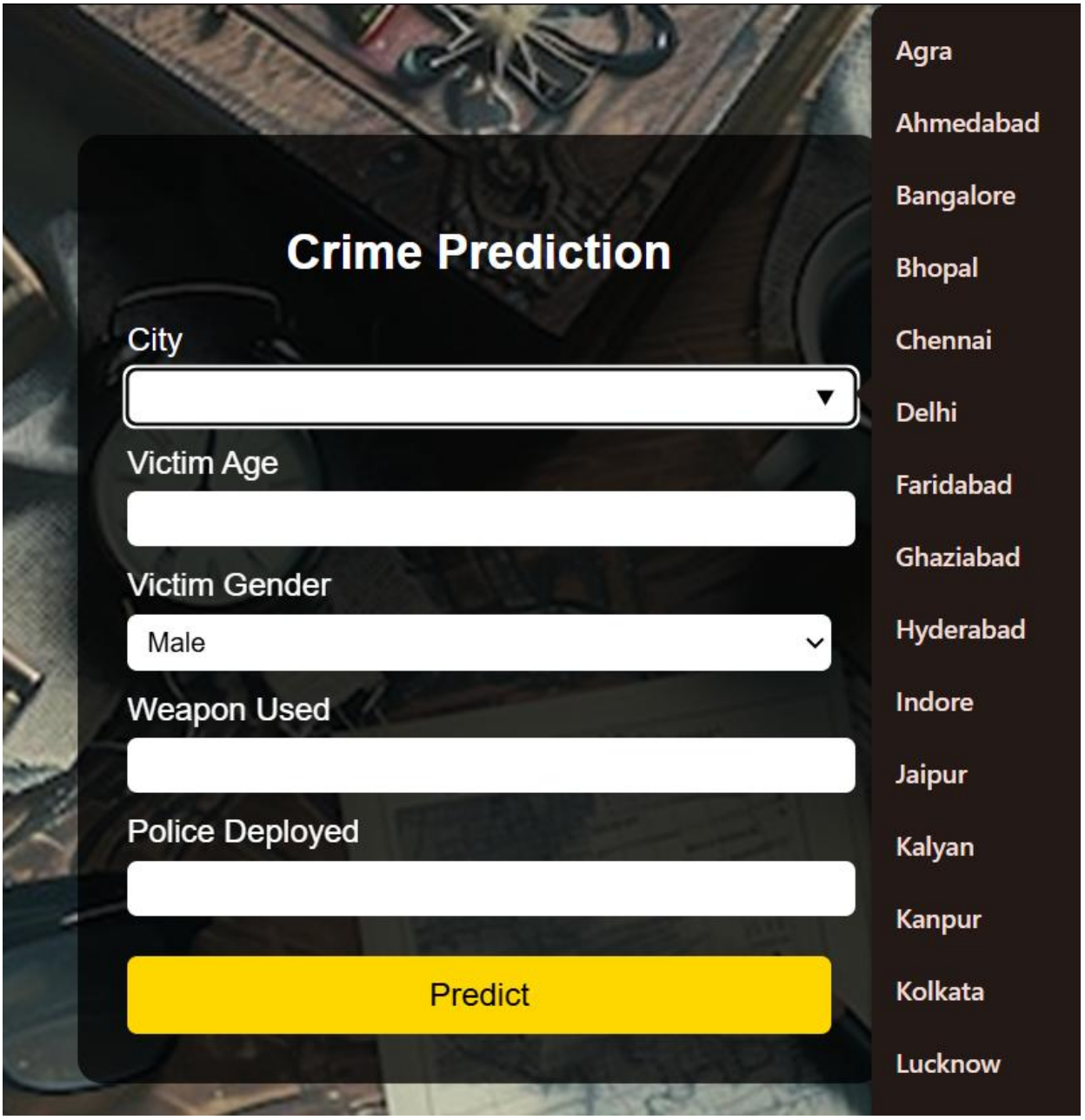


Fig.8.3: Shows the Crime Rate Browsing Portal of our pro

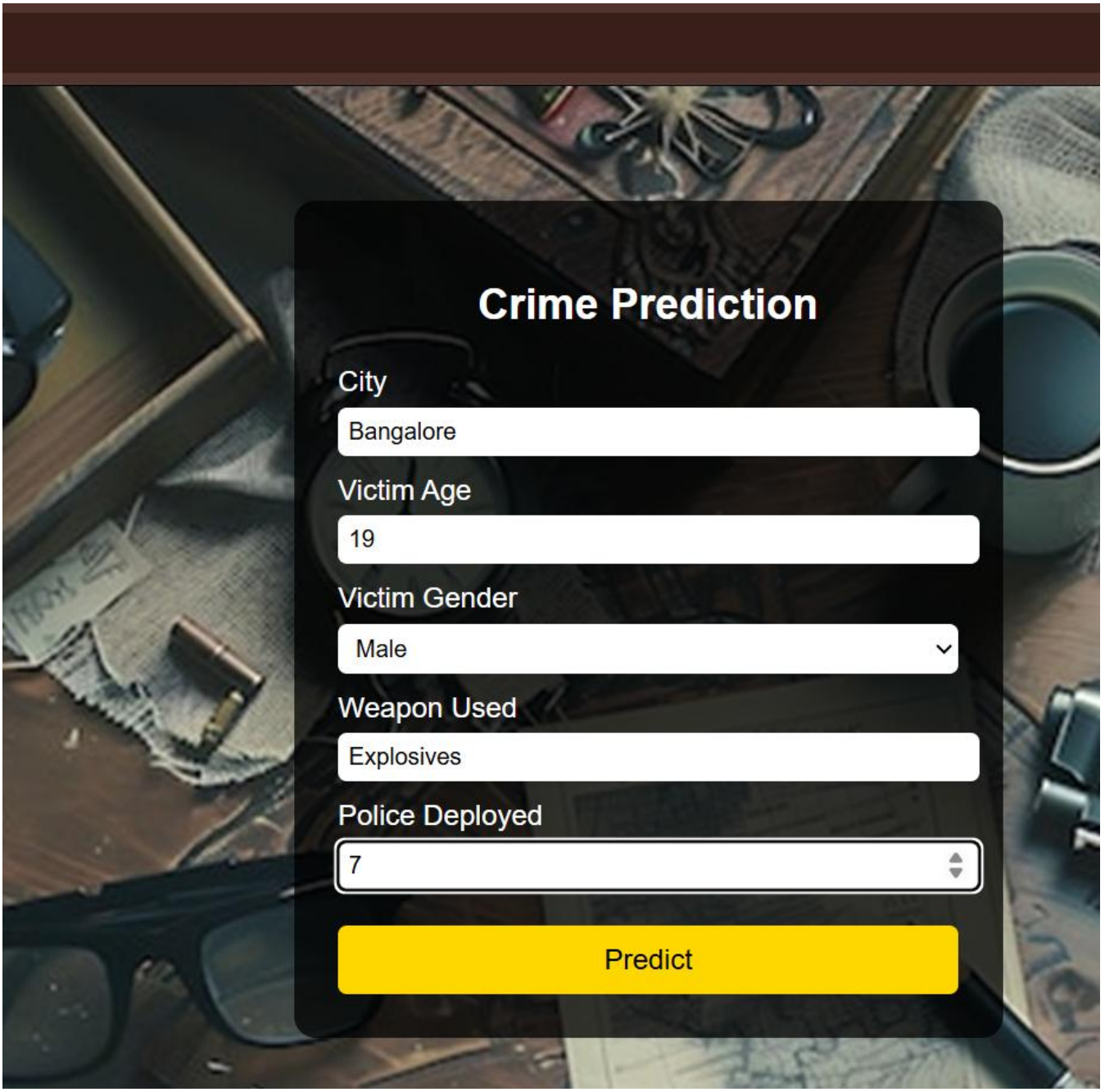


Fig.8.4: Shows the web page of filling the details about crime location, victim age,victim gender,Weapon used predict the crime.

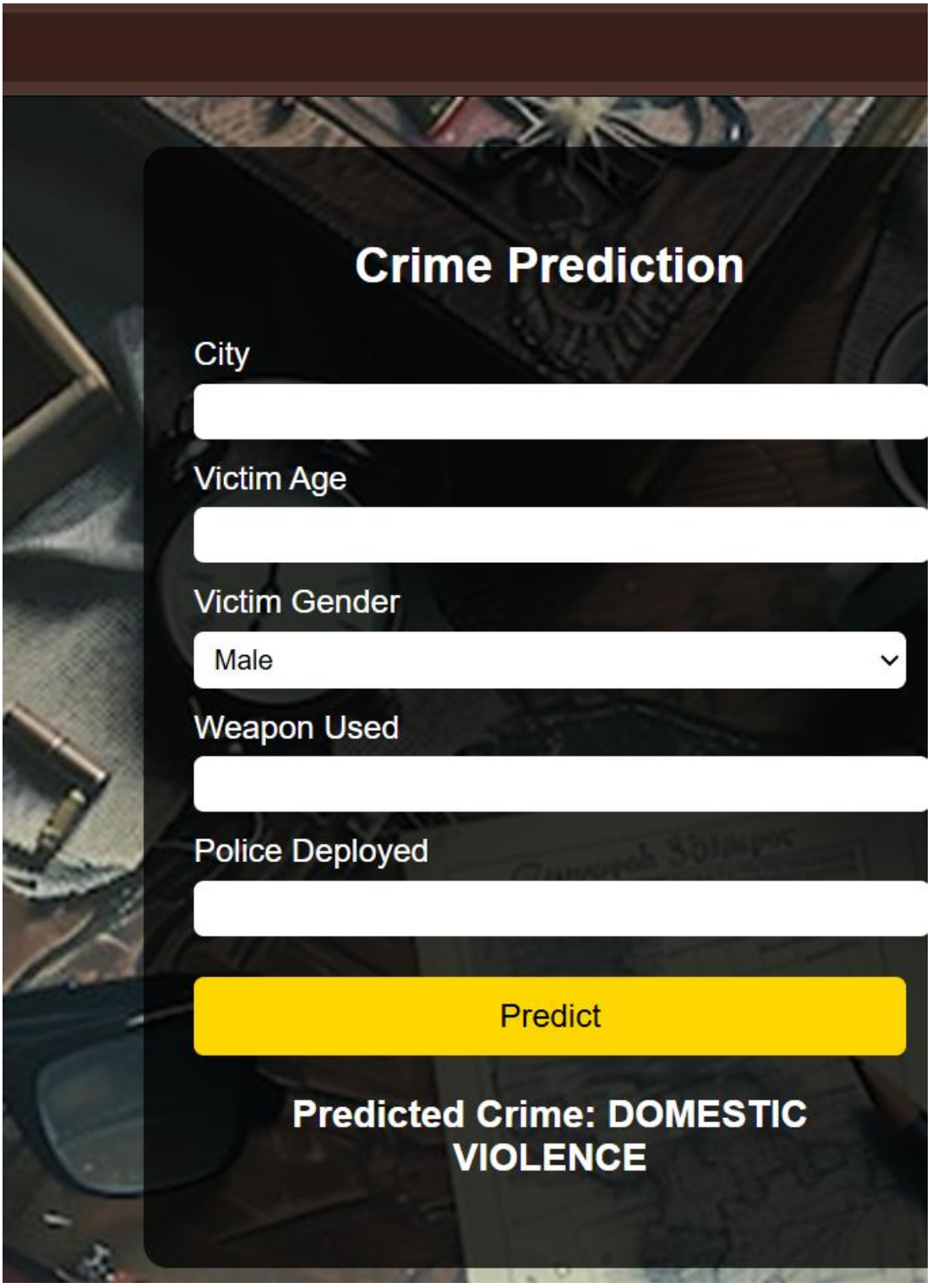


Fig.8.5: Shows the prediction of type of crime.

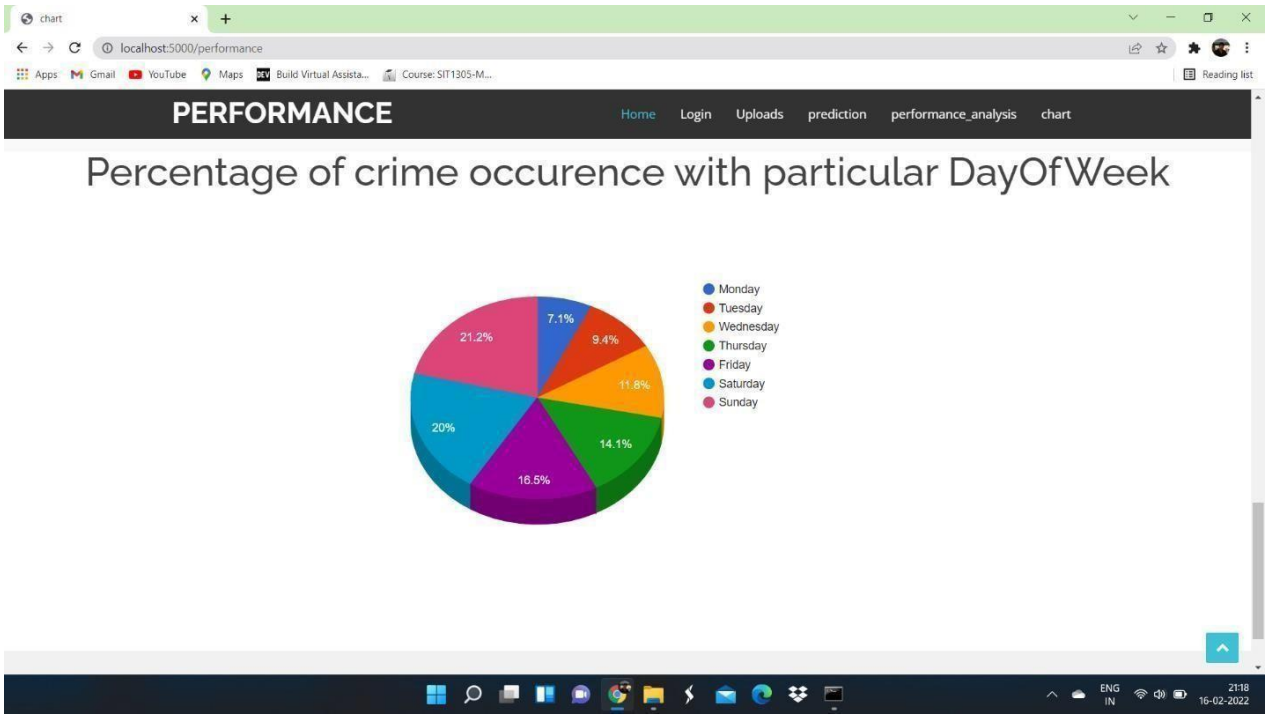


Fig8.6: Shows the performance of crime occurrences with particular day of Week

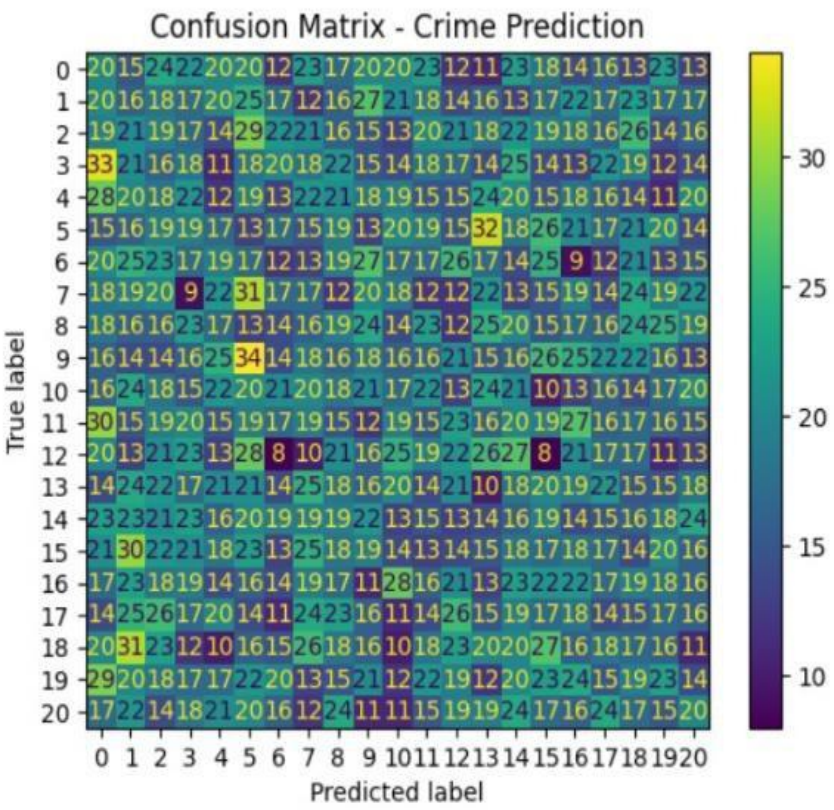


Fig.8.7: Confusion Matrix

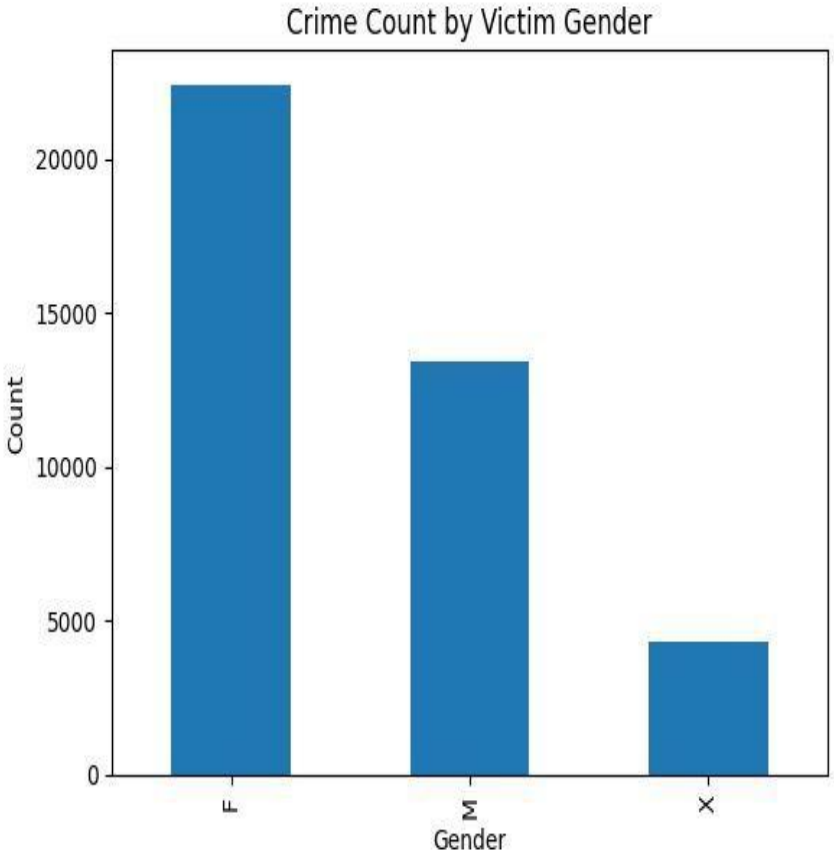


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Fig.8.8: Shows the graph for crime occurrences with particular victim gender

CHAPTER – 8

SYSTEM TESTING

8.1. TESTING OF PRODUCT

System testing is the process of verifying and validating the complete working of the application to ensure that it meets the specified requirements. In this project, system testing is performed to check whether the crime prediction system works correctly from input to output.

The entire system, including data input, pre-processing, model prediction, and output display through the Flask web interface, is tested to ensure accuracy, reliability, and proper functionality.

8.2. TYPES OF TESTING

- **Unit testing**

Unit testing involves the design of test cases that validate that the internal program logic is functioning properly, and that program inputs produce valid outputs. All decision branches and internal code flow should be validated. It is the testing of individual software units of the application .it is done after the completion of an individual unit before integration. This is a structural testing, that relies on knowledge of its construction and is invasive. Unit tests perform basic tests at component level and test a specific business process, application, and/or system configuration. Unit tests ensure that each unique path of a business process performs accurately to the documented specifications and contains clearly defined inputs and expected results.

- **Integration testing**

Integration tests are designed to test integrated software components to determine if they actually run as one program. Testing is event driven and is more concerned with the basic outcome of screens or fields. Integration tests

demonstrate that although the components were individually satisfaction, as shown by successfully unit testing, the combination of components is correct and consistent. Integration testing is specifically aimed at exposing the problems that arise from the combination of components.

➢ Functional test

Functional tests provide systematic demonstrations that functions tested are available as specified by the business and technical requirements, system documentation, and user manuals.

Functional testing is centered on the following items:

Valid Input: identified classes of valid input must be accepted.
Invalid Input: identified classes of invalid input must be rejected.
Functions: identified functions must be exercised.
Output: identified classes of application outputs must be exercised.
Systems/Procedures: interfacing systems or procedures must be invoked.

Organization and preparation of functional tests is focused on requirements, key functions, or special test cases. In addition, systematic coverage pertaining to identify Business process flows; data fields, predefined processes, and successive processes must be considered for testing. Before functional testing is complete, additional tests are identified and the effective value of current tests is determined.

➢ System Test

System testing ensures that the entire integrated software system meets requirements. It tests a configuration to ensure known and predictable results. An example of system testing is the configuration-oriented system integration test. System testing is based on process descriptions and flows, emphasizing pre-driven process links and integration points.

8.3. Test strategy and approach

The test strategy defines how testing is carried out in the project.
In this project, the strategy is to ensure that the **crime prediction system works accurately, efficiently, and reliably**.

- Testing is done at different levels: unit, integration, and system testing
- Both **functional and performance testing** are included
- Real dataset inputs are used to check model accuracy
- The Flask web application is tested for correct user interaction

Features to be tested

- Verify that the entries are of the correct format
- No duplicate entries should be allowed
- All links should take the user to the correct page.

Test Approach

The test approach describes the step-by-step method followed for testing.

1. **Module Testing:**

   Each module like data preprocessing, feature selection, and model prediction is tested separately.

2. **Integration Testing:**

   All modules are combined and tested to ensure smooth data flow

3. **System Testing:**

   The complete application is tested from user input to final prediction

4. **Validation of Results:**

   Model predictions are verified using evaluation metrics like accuracy and confusion matrix

CHAPTER- 9

9.1 CONCLUSION:

This project successfully demonstrates the use of machine learning techniques for crime rate analysis and prediction. By analyzing real-world crime data, the system is able to identify patterns and predict the type of crime based on given inputs. Techniques such as data preprocessing, feature selection, and model training help improve the accuracy and efficiency of the system.

Among the models used, Random Forest provides better performance compared to other algorithms. The integration of a Flask-based web application makes the system user-friendly and allows real-time predictions. Visualization tools further help in understanding crime trends and patterns effectively.

Overall, this project provides a useful solution for crime analysis, helping law enforcement agencies in decision-making and contributing to improved public safety.

9.2 FUTURE ENHANCEMENT:

This project can be further improved by incorporating advanced features and technologies to enhance accuracy and usability.

- Integration of **real-time crime data** from police databases and surveillance systems
- Implementation of **advanced machine learning and deep learning models** for better prediction accuracy
- Use of **GIS-based mapping** to visualize crime hotspots geographically .
- Development of a **mobile application** for easy access and wider usage.

- Integration of **alert and notification systems** to inform authorities instantly.
- Deployment on **cloud platforms** for scalability and large-scale data analysis.
- Inclusion of **more features** like weather, time patterns, and socio-economic factors for improved prediction

SAMPLE CODING:

PYTHON(app.py):

```
from flask import Flask, render_template, request
import pickle

app = Flask(__name__)

model = pickle.load(open('../model/model.pkl', 'rb'))
encoders = pickle.load(open('../model/encoders.pkl', 'rb'))
label_encoder = pickle.load(open('../model/label_encoder.pkl', 'rb'))

# Get suggestions from trained encoders
cities = sorted(
    [str(c) for c in encoders['City'].classes_ if str(c) != 'nan']
)
weapons = sorted(
    [str(w) for w in encoders['Weapon Used'].classes_ if str(w) != 'nan']
)


@app.route('/', methods=['GET', 'POST'])
```

```
def index():
    prediction = None

    if request.method == 'POST':
        city = request.form['city']
        gender = request.form['gender']   # M / F / O
        weapon = request.form['weapon']
        age = int(request.form['age'])
        police = int(request.form['police'])

        data = [
            encoders['City'].transform([city])[0],
            age,
            encoders['Victim Gender'].transform([gender])[0],
            encoders['Weapon Used'].transform([weapon])[0],
            police
        ]

        pred = model.predict([data])
        prediction = label_encoder.inverse_transform(pred)[0]

    return render_template(
        'index.html',
        prediction=prediction,
        cities=cities,
        weapons=weapons
    )

if __name__ == '__main__':
    app.run(debug=True)
```

HTML:

```
<!DOCTYPE html>
<html>
<head>
    <title>Crime Rate Prediction</title>

    <style>
        body {
            background-image: url("{{ url_for('static', filename='bg.jpg') }}");
            background-size: cover;
            background-repeat: no-repeat;
            font-family: Arial, sans-serif;
            color: white;
        }

        .container {
            display: flex;
            justify-content: center;
            align-items: center;
            height: 100vh;
        }
```

```
form {
    background: rgba(0,0,0,0.75);
    padding: 25px;
    width: 350px;
    border-radius: 12px;
}

label {
    display: block;
    margin-top: 10px;
}

input, select {
    width: 100%;
    padding: 6px;
    margin-top: 5px;
    border-radius: 5px;
    border: none;
}

button {
    margin-top: 20px;
    width: 100%;
    padding: 10px;
    background: gold;
    border: none;
    border-radius: 5px;
    font-size: 16px;
}
```

```
        h2, h3 {
            text-align: center;
        }
    </style>
</head>

<body>

<div class="container">
<form method="post">

    <h2>Crime Prediction</h2>

    <label>City</label>
    <input list="cityList" name="city" required>
    <datalist id="cityList">
        {% for city in cities %}
        <option value="{{ city }}">
        {% endfor %}
    </datalist>

    <label>Victim Age</label>
    <input type="number" name="age" required>

    <label>Victim Gender</label>
    <select name="gender" required>
        <option value="M">Male</option>
        <option value="F">Female</option>
        <option value="O">Other</option>
```

```
    </select>

    <label>Weapon Used</label>
    <input list="weaponList" name="weapon" required>
    <datalist id="weaponList">
        {% for weapon in weapons %}
        <option value="{{ weapon }}">
        {% endfor %}
    </datalist>

    <label>Police Deployed</label>
    <input type="number" name="police" required>

    <button type="submit">Predict</button>

    {% if prediction %}
    <h3>Predicted Crime: {{ prediction }}</h3>
    {% endif %}

</form>
</div>

</body>
</html>
```

TRAINING THE MODEL:

```
import pandas as pd
import pickle
from sklearn.preprocessing import LabelEncoder
from sklearn.ensemble import RandomForestClassifier
```

```
# Load dataset
df = pd.read_csv("data/crime_dataset_india.csv")

# Select features and target
features = ["City", "Victim Age", "Victim Gender", "Weapon Used", "Police Deployed"]
X = df[features]
y = df["Crime Description"]

# Encode categorical features
encoders = {}
for col in X.select_dtypes(include="object"):
    le = LabelEncoder()
    X[col] = le.fit_transform(X[col])
    encoders[col] = le

label_encoder = LabelEncoder()
y_encoded = label_encoder.fit_transform(y)

# Train model
model = RandomForestClassifier(n_estimators=80, random_state=42)
model.fit(X, y_encoded)

# Save model and encoders
pickle.dump(model, open("model/model.pkl", "wb"))
pickle.dump(encoders, open("model/encoders.pkl", "wb"))
pickle.dump(label_encoder, open("model/label_encoder.pkl", "wb"))

print("✅ Model trained successfully on REAL dataset")
```

REFERENCES

[1] Abdul Jalil, M. M., Mohd, F., & Mohamad Noor, N. M. (2017). A comparative study
to evaluate filtering methods for crime data feature selection. Procedia Computer Science.

[2] Al Boni, M., & Gerber, M. S. (2017). Area-specific crime prediction models. In Proceedings - 2016 15th IEEE international conference on machine learning and applications, ICMLA 2016 (pp. 671–676). https://doi.org/10.1109/ICMLA.2016.43

[3] Albo, F.. A survey of research into artificial neural networks for crime prediction. n.d, Retrieved May 21, 2021, from
https://www.researchgate.net/publication/336116157.

[4] Aldossari, B. S., Alqahtani, F. M., Alshahrani, N. S., Alhammam, M. M., Alzamanan,
R. M., Aslam, N., & Irfanullah. (2020). A comparative study of decision tree and naive bayes machine learning model for crime category prediction in chicago. ACM International Conference Proceeding Series, 34–38. https://doi.org/ 10.1145/3379247.3379279

[5] Alkhaibari, A. A., & Chung, P. T. (2017). Cluster analysis for reducing city crime
rates. In 2017 IEEE long island systems, applications and technology conference. https://doi.org/ 10.1109/LISAT.2017.8001983. LISAT 2017.

[6] Almaw, A., & Kadam, K. (2019). Crime data analysis and prediction using ensemble
learning. In Proceedings of the 2nd international conference on intelligent computing and control systems, ICICCS 2018 (pp. 1918–1923). Iciccs. https://doi.org/10.1109/ ICCONS.2018.8663186.59

[7] Altameem, T., & Amoon, M. (2019). Crime activities prediction using hybridization
of f irefly optimization technique and fuzzy cognitive map neural networks. Neural Computing & Applications, 31(5), 1263–1273. https://doi.org/10.1007/s00521-018-3561-7

[8] Alves, L. G. A., Ribeiro, H. V., & Rodrigues, F. A. (2018). Crime prediction through
urban metrics and statistical learning. Physica A: Statistical Mechanics and its Applications, 505, 435–443. https://doi.org/10.1016/j.physa.2018.03.084

[9] Anuar, S., Selamat, A., & Sallehuddin, R. (2015). Hybrid artificial neural network
with artificial bee colony algorithm for crime classification. Advances in Intelligent Systems and Computing, 331, 31–40. https://doi.org/10.1007/978-3-319-13153-5_4

[10] Azeez, J., & Aravindhar, D. J. (2015). Hybrid approach to crime prediction using deep learning. In 2015 international conference on advances in computing, communications and informatics, ICACCI 2015 (pp. 1701–1710). https://doi.org/10.1109/ ICACCI.2015.7275858

[11] Babakura, A., Sulaiman, M. N., & Yusuf, M. A. (2015). Improved method of classification algorithms for crime prediction. In Proceedings - 2014 international symposium on biometrics and security technologies, ISBAST 2014 (pp. 250–255). https://doi.org/ 10.1109/ISBAST.2014.7013130

[12] Baculo, M. J. C., Marzan, C. S., De Dios Bulos, R., & Ruiz, C. (2017). Geospatial-temporal analysis andclassification of criminal data in Manila. In 2017 2nd IEEE international conference on computational intelligence and applications, ICCIA 2017, 2017-Janua, 6–11. https://doi.org/10.1109/CIAPP.2017.8167050

[13] Bandekar, S. R., & Vijayalakshmi, C. (2020). Design and analysis of machine learning algorithms for the reduction of crime rates in India. Procedia Computer60 Science, 172, 122–127. https://doi.org/10.1016/j.procs.2020.05.018

[14] Barreira, R., Pinheiro, V., & Furtado, V. (2017). A framework for digital forensics analysis based on semantic role labeling. In 2017 IEEE international conference on intelligence and security informatics: Security and big data, ISI 2017 (pp. 66–71). https://doi.org/ 10.1109/ISI.2017.8004876

[15] Belesiotis, A., Papadakis, G., & Skoutas, D. (2018). Analyzing and predicting spatial crime distribution using crowdsourced and open data. ACM Transactions on Spatial Algorithms and Systems, 3(4). https://doi.org/10.1145/3190345

[16] Bellarmine, A. B. (2023). Identity using machine learning approach. In 2023 2nd international conference on trends in electronics and informatics (ICOEI), icoei (pp. 107–114).

[17] Bharathi, S. T., Indrani, B., & Prabakar, M. A. (2018). A supervised learning approach for criminal identification using similarity measures and K-Medoids clustering. In 2017 international conference on intelligent computing, instrumentation and control technologies, ICICICT 2017, 2018-Janua (pp. 646–653). https://doi.org/10.1109/ ICICICT1.2017.8342639

[18] Biswas, A. A., & Basak, S. (2022). Forecasting the trends and patterns of crime in Bangladesh using machine learning model. In 2022 2nd international conference on intelligent communication and computational techniques, ICCT 2022 (pp. 114– . https://doi.org/10.1109/ICCT46177.2019.8969031.